

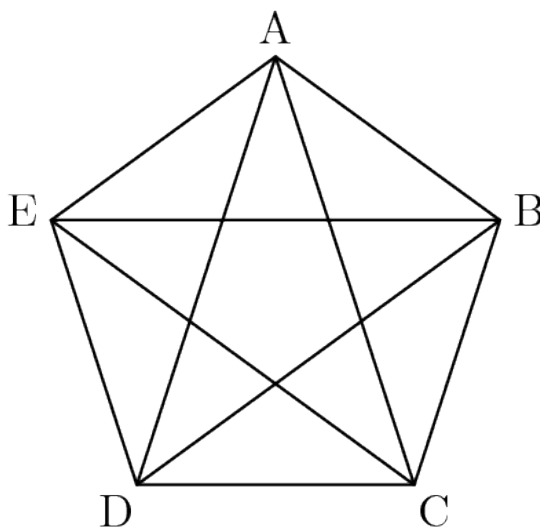
USAMO

§1 USAMO 1972

1. (*USAMO 1972*) The symbols $(a, b, \dots, g)$ and $[a, b, \dots, g]$ denote the greatest common divisor and least common multiple, respectively, of the positive integers $a, b, \dots, g$. For example, $(3, 6, 18) = 3$ and $[6, 15] = 30$. Prove that

$$\frac{[a, b, c]^2}{[a, b][b, c][c, a]} = \frac{(a, b, c)^2}{(a, b)(b, c)(c, a)}.$$

2. (*USAMO 1972*) A given tetrahedron $ABCD$ is isosceles, that is, $AB = CD$, $AC = BD$, $AD = BC$. Show that the faces of the tetrahedron are acute-angled triangles.
3. (*USAMO 1972*) A random number selector can only select one of the nine integers 1, 2, ..., 9, and it makes these selections with equal probability. Determine the probability that after n selections ($n > 1$), the product of the n numbers selected will be divisible by 10.
4. (*USAMO 1972*) Let R denote a non-negative rational number. Determine a fixed set of integers a, b, c, d, e, f , such that for *every* choice of R ,
- $$\left| \frac{aR^2 + bR + c}{dR^2 + eR + f} - \sqrt[3]{2} \right| < |R - \sqrt[3]{2}|$$
5. (*USAMO 1972*) A given convex pentagon $ABCDE$ has the property that the area of each of the five triangles ABC , BCD , CDE , DEA , and EAB is unity. Show that all pentagons with the above property have the same area, and calculate that area. Show, furthermore, that there are infinitely many non-congruent pentagons having the above area property.



§2 USAMO 1973

1. (USAMO 1973) Two points P and Q lie in the interior of a regular tetrahedron $ABCD$. Prove that angle $PAQ < 60^\circ$.

2. (USAMO 1973) Let $\{X_n\}$ and $\{Y_n\}$ denote two sequences of integers defined as follows:

$$X_0 = 1, X_1 = 1, X_{n+1} = X_n + 2X_{n-1} \quad (n = 1, 2, 3, \dots),$$

$$Y_0 = 1, Y_1 = 7, Y_{n+1} = 2Y_n + 3Y_{n-1} \quad (n = 1, 2, 3, \dots).$$

Thus, the first few terms of the sequences are:

$$X : 1, 1, 3, 5, 11, 21, \dots,$$

$$Y : 1, 7, 17, 55, 161, 487, \dots$$

Prove that, except for the "1", there is no term which occurs in both sequences.

3. (USAMO 1973) Three distinct vertices are chosen at random from the vertices of a given regular polygon of $(2n + 1)$ sides. If all such choices are equally likely, what is the probability that the center of the given polygon lies in the interior of the triangle determined by the three chosen random points?

4. (USAMO 1973) Determine all the **roots, real or complex**, of the system of simultaneous **equations**

$$x + y + z = 3, \quad x^2 + y^2 + z^2 = 3,$$

$$x^3 + y^3 + z^3 = 3.$$

5. (USAMO 1973) Show that the cube roots of three distinct prime numbers cannot be three terms (not necessarily consecutive) of an arithmetic progression.

§3 USAMO 1974

1. (USAMO 1974) Let a , b , and c denote three distinct integers, and let P denote a polynomial having all integral coefficients. Show that it is impossible that $P(a) = b$, $P(b) = c$, and $P(c) = a$.

2. (USAMO 1974) Prove that if a , b , and c are positive real numbers, then

$$a^a b^b c^c \geq (abc)^{(a+b+c)/3}$$

3. (USAMO 1974) Two boundary points of a ball of radius 1 are joined by a curve contained in the ball and having length less than 2. Prove that the curve is contained entirely within some hemisphere of the given ball.

4. (USAMO 1974) A father, mother and son hold a family tournament, playing a two person board game with no ties. The tournament rules are:

(i) The weakest player chooses the first two contestants.

(ii) The winner of any game plays the next game against the person left out.

(iii) The first person to win two games wins the tournament.

The father is the weakest player, the son the strongest, and it is assumed that any player's probability of winning an individual game from another player does not change during the tournament. Prove that the father's optimal strategy for winning the tournament is to play the first game with his wife.

5. (USAMO 1974) Consider the two triangles $\triangle ABC$ and $\triangle PQR$ shown in Figure 1. In $\triangle ABC$, $\angle ADB = \angle BDC = \angle CDA = 120^\circ$. Prove that $x = u + v + w$.

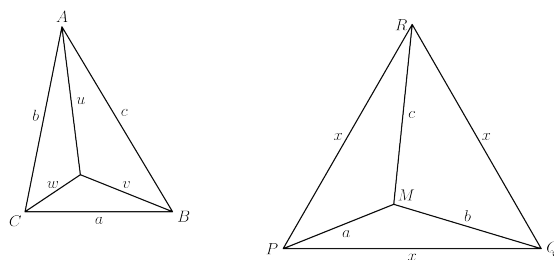


Figure 1

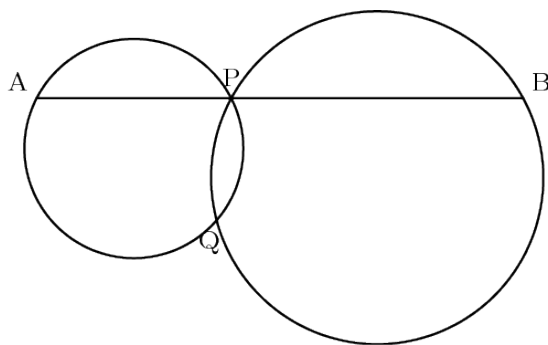
§4 USAMO 1975

- (USAMO 1975) (a) Prove that

$$[5x] + [5y] \geq [3x + y] + [3y + x],$$
 where $x, y \geq 0$ and $[u]$ denotes the greatest integer $\leq u$ (e.g., $[\sqrt{2}] = 1$).
 (b) Using (a) or otherwise, prove that

$$\frac{(5m)!(5n)!}{m!n!(3m+n)!(3n+m)!}$$
 is integral for all positive integral m and n .
- (USAMO 1975) Let A, B, C, D denote four points in space and AB the distance between A and B , and so on. Show that

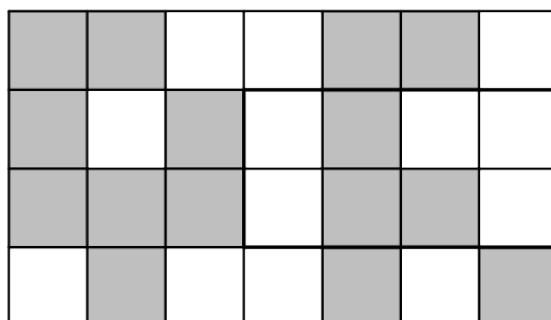
$$AC^2 + BD^2 + AD^2 + BC^2 \geq AB^2 + CD^2.$$
- (USAMO 1975) If $P(x)$ denotes a polynomial of degree n such that $P(k) = k/(k+1)$ for $k = 0, 1, 2, \dots, n$, determine $P(n+1)$.
- (USAMO 1975) Two given circles intersect in two points P and Q . Show how to construct a segment AB passing through P and terminating on the two circles such that $AP \cdot PB$ is a maximum.



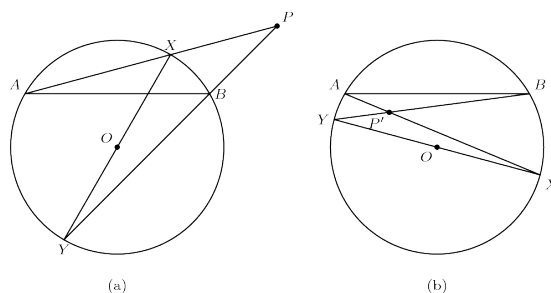
- (USAMO 1975) A deck of n playing cards, which contains three aces, is shuffled at random (it is assumed that all possible card distributions are equally likely). The cards are then turned up one by one from the top until the second ace appears. Prove that the expected (average) number of cards to be turned up is $(n+1)/2$.

§5 USAMO 1976

1. (USAMO 1976)



- (a) Suppose that each square of a 4×7 chessboard, as shown above, is colored either black or white. Prove that with *any* such coloring, the board must contain a rectangle (formed by the horizontal and vertical lines of the board such as the one outlined in the figure) whose four distinct unit corner squares are all of the same color.
 - (b) Exhibit a black-white coloring of a 4×6 board in which the four corner squares of every rectangle, as described above, are not all of the same color.
2. (USAMO 1976) If A and B are fixed points on a given circle and XY is a variable diameter of the same circle, determine the locus of the point of intersection of lines AX and BY . You may assume that AB is not a diameter.



3. (USAMO 1976) Determine all integral solutions of $a^2 + b^2 + c^2 = a^2b^2$.
4. (USAMO 1976) If the sum of the lengths of the six edges of a trirectangular tetrahedron $PABC$ (i.e., $\angle APB = \angle BPC = \angle CPA = 90^\circ$) is S , determine its maximum volume.
5. (USAMO 1976) If $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ are all **polynomials** such that

$$P(x^5) + xQ(x^5) + x^2R(x^5) = (x^4 + x^3 + x^2 + x + 1)S(x),$$

prove that $x - 1$ is a factor of $P(x)$.

§6 USAMO 1977

1. (USAMO 1977) Determine all pairs of positive integers (m, n) such that $(1 + x^n + x^{2n} + \cdots + x^{mn})$ is divisible by $(1 + x + x^2 + \cdots + x^m)$.

2. (USAMO 1977) ABC and $A'B'C'$ are two triangles in the same plane such that the lines AA' , BB' , CC' are mutually parallel. Let $[ABC]$ denote the area of triangle ABC with an appropriate $\pm$ sign, etc.; prove that

$$3([ABC] + [A'B'C']) = [AB'C'] + [BC'A'] + [CA'B'] + [A'BC] + [B'CA] + [C'AB].$$

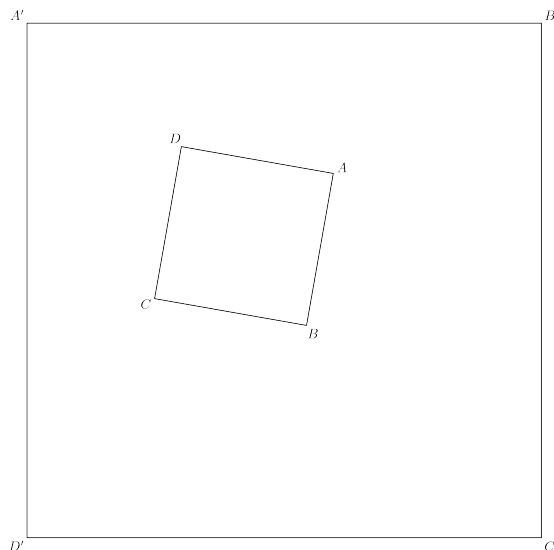
3. (USAMO 1977) If a and b are two of the roots of $x^4 + x^3 - 1 = 0$, prove that ab is a root of $x^6 + x^4 + x^3 - x^2 - 1 = 0$.
4. (USAMO 1977) Prove that if the opposite sides of a skew (non-planar) quadrilateral are congruent, then the line joining the midpoints of the two diagonals is perpendicular to these diagonals, and conversely, if the line joining the midpoints of the two diagonals of a skew quadrilateral is perpendicular to these diagonals, then the opposite sides of the quadrilateral are congruent.
5. (USAMO 1977) If a, b, c, d, e are positive numbers bounded by p and q , i.e, if they lie in $[p, q]$, $0 < p$, prove that

$$(a + b + c + d + e) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} + \frac{1}{e} \right) \leq 25 + 6 \left(\sqrt{\frac{p}{q}} - \sqrt{\frac{q}{p}} \right)^2$$

and determine when there is equality.

§7 USAMO 1978

1. (USAMO 1978) Given that a, b, c, d, e are real numbers such that
- $$a + b + c + d + e = 8,$$
- $$a^2 + b^2 + c^2 + d^2 + e^2 = 16.$$
- Determine the maximum value of e .
2. (USAMO 1978) $ABCD$ and $A'B'C'D'$ are square maps of the same region, drawn to different scales and superimposed as shown in the figure. Prove that there is only one point O on the small map that lies directly over point O' of the large map such that O and O' each represent the same place of the country. Also, give a Euclidean construction (straight edge and compass) for O .



3. (*USAMO 1978*) An integer n will be called *good* if we can write
- $$n = a_1 + a_2 + \cdots + a_k,$$
- where $a_1, a_2, \dots, a_k$ are positive integers (not necessarily distinct) satisfying
- $$\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_k} = 1.$$
- Given the information that the integers 33 through 73 are good, prove that every integer ≥ 33 is good.
4. (*USAMO 1978*) (a) Prove that if the six dihedral (i.e. angles between pairs of faces) of a given tetrahedron are congruent, then the tetrahedron is regular.
- (b) Is a tetrahedron necessarily regular if five dihedral angles are congruent?
5. (*USAMO 1978*) Nine mathematicians meet at an international conference and discover that among any three of them, at least two speak a common language. If each of the mathematicians speak at most three languages, prove that there are at least three of the mathematicians who can speak the same language.

§8 USAMO 1979

1. (*USAMO 1979*) Determine all non-negative integral solutions $(n_1, n_2, \dots, n_{14})$ if any, apart from permutations, of the Diophantine Equation $n_1^4 + n_2^4 + \cdots + n_{14}^4 = 1599$.
2. (*USAMO 1979*) N is the north pole. A and B are points on a great circle through N equidistant from N . C is a point on the equator. Show that the great circle through C and N bisects the angle ACB in the spherical triangle ABC (a spherical triangle has great circle arcs as sides).
3. (*USAMO 1979*) $a_1, a_2, \dots, a_n$ is an arbitrary sequence of positive integers. A member of the sequence is picked at random. Its value is a . Another member is picked at random, independently of the first. Its value is b . Then a third value, c . Show that the probability that $a + b + c$ is divisible by 3 is at least $\frac{1}{4}$.
4. (*USAMO 1979*) P lies between the rays OA and OB . Find Q on OA and R on OB collinear with P so that $\frac{1}{PQ} + \frac{1}{PR}$ is as large as possible.
5. (*USAMO 1979*) Let $A_1, A_2, \dots, A_{n+1}$ be distinct subsets of $[n]$ with $|A_1| = |A_2| = \cdots = |A_{n+1}| = 3$. Prove that $|A_i \cap A_j| = 1$ for some pair $\{i, j\}$. Note that $[n] = \{1, 2, 3, \dots, n\}$, or, alternatively, $\{x : 1 \leq x \leq n\}$.

§9 USAMO 1980

1. (*USAMO 1980*) A balance has unequal arms and pans of unequal weight. It is used to weigh three objects. The first object balances against a weight A , when placed in the left pan and against a weight a , when placed in the right pan. The corresponding weights for the second object are B and b . The third object balances against a weight C , when placed in the left pan. What is its true weight?
2. (*USAMO 1980*) Find the maximum possible number of three term arithmetic progressions in a monotone sequence of n distinct reals.
3. (*USAMO 1980*) $A + B + C$ is an integral multiple of π . x, y , and z are real numbers. If $x \sin(A) + y \sin(B) + z \sin(C) = x^2 \sin(2A) + y^2 \sin(2B) + z^2 \sin(2C) = 0$, show that $x^n \sin(nA) + y^n \sin(nB) + z^n \sin(nC) = 0$ for any positive integer n .

4. (USAMO 1980) The insphere of a tetrahedron touches each face at its centroid. Show that the tetrahedron is regular.
5. (USAMO 1980) If x, y, z are reals such that $0 \leq x, y, z \leq 1$, show that $\frac{x}{y+z+1} + \frac{y}{z+x+1} + \frac{z}{x+y+1} \leq 1 - (1-x)(1-y)(1-z)$

§10 USAMO 1981

1. (USAMO 1981) Prove that if n is not a multiple of 3, then the angle $\frac{\pi}{n}$ can be trisected with ruler and compasses.
2. (USAMO 1981) What is the largest number of towns that can meet the following criteria. Each pair is directly linked by just one of air, bus or train. At least one pair is linked by air, at least one pair by bus and at least one pair by train. No town has an air link, a bus link and a train link. No three towns, A, B, C are such that the links between AB, AC and BC are all air, all bus or all train.
3. (USAMO 1981) Show that for any triangle, $\frac{3\sqrt{3}}{2} \geq \sin(3A) + \sin(3B) + \sin(3C) \geq -2$. When does the equality hold?
4. (USAMO 1981) The sum of the measures of all the face angles of a given complex polyhedral angle is equal to the sum of all its dihedral angles. Prove that the polyhedral angle is a trihedral angle.
5. (USAMO 1981) Show that for any positive real x ,

$$[nx] \geq \sum_{k=1}^n \left(\frac{[kx]}{k} \right),$$

where $[x]$ denotes the greatest integer not exceeding x .

§11 USAMO 1982

1. (USAMO 1982) In a party with 1982 persons, among any group of four there is at least one person who knows each of the other three. What is the minimum number of people in the party who know everyone else?
2. (USAMO 1982) Let $S_r = x^r + y^r + z^r$ with x, y, z real. It is known that if $S_1 = 0$,
 $(*) \frac{S_{m+n}}{m+n} = \frac{S_m}{m} \frac{S_n}{n}$
for $(m, n) = (2, 3), (3, 2), (2, 5),$ or $(5, 2)$. Determine *all* other pairs of integers (m, n) if any, so that $(*)$ holds for all real numbers x, y, z such that $S_1 = 0$.
3. (USAMO 1982) If a point A_1 is in the interior of an equilateral triangle ABC and point A_2 is in the interior of $\triangle A_1BC$, prove that
 $I.Q.(A_1BC) > I.Q.(A_2BC),$
where the *isoperimetric quotient* of a figure F is defined by

$$I.Q.(F) = \frac{\text{Area}(F)}{[\text{Perimeter}(F)]^2}$$
4. (USAMO 1982) Prove that there exists a positive integer k such that $k \cdot 2^n + 1$ is composite for every positive integer n .

5. (*USAMO 1982*) A, B , and C are three interior points of a sphere S such that AB and AC are perpendicular to the diameter of S through A , and so that two spheres can be constructed through A, B , and C which are both tangent to S . Prove that the sum of their radii is equal to the radius of S .

§12 USAMO 1983

1. (*USAMO 1983*) On a given circle, six points A, B, C, D, E , and F are chosen at random, independently and uniformly with respect to arc length. Determine the probability that the two triangles ABC and DEF are disjoint, i.e., have no common points.
2. (*USAMO 1983*) Prove that the roots of

$$x^5 + ax^4 + bx^3 + cx^2 + dx + e = 0$$

cannot all be real if $2a^2 < 5b$.

3. (*USAMO 1983*) Each set of a finite family of subsets of a line is a union of two closed intervals. Moreover, any three of the sets of the family have a point in common. Prove that there is a point which is common to at least half the sets of the family.
4. (*USAMO 1983*) Six segments S_1, S_2, S_3, S_4, S_5 , and S_6 are given in a plane. These are congruent to the edges AB, AC, AD, BC, BD , and CD , respectively, of a tetrahedron $ABCD$. Show how to construct a segment congruent to the altitude of the tetrahedron from vertex A with straight-edge and compasses.
5. (*USAMO 1983*) Consider an open interval of length $1/n$ on the real number line, where n is a positive integer. Prove that the number of irreducible fractions p/q , with $1 \leq q \leq n$, contained in the given interval is at most $(n+1)/2$.

§13 USAMO 1984

1. (*USAMO 1984*) The product of two of the four roots of the quartic equation $x^4 - 18x^3 + kx^2 + 200x - 1984 = 0$ is -32 . Determine the value of k .
2. (*USAMO 1984*) The geometric mean of any set of m non-negative numbers is the m -th root of their product.
- (i) For which positive integers n is there a finite set S_n of n distinct positive integers such that the geometric mean of any subset of S_n is an integer?
- (ii) Is there an infinite set S of distinct positive integers such that the geometric mean of any finite subset of S is an integer?
3. (*USAMO 1984*) P, A, B, C , and D are five distinct points in space such that $\angle APB = \angle BPC = \angle CPD = \angle DPA = \theta$, where θ is a given acute angle. Determine the greatest and least values of $\angle APC + \angle BPD$.
4. (*USAMO 1984*) A difficult mathematical competition consisted of a Part I and a Part II with a combined total of 28 problems. Each contestant solved 7 problems altogether. For each pair of problems, there were exactly two contestants who solved both of them. Prove that there was a contestant who, in Part I, solved either no problems or at least four problems.

5. (*USAMO 1984*) $P(x)$ is a polynomial of degree $3n$ such that

$$\begin{aligned} P(0) = P(3) = \cdots &= P(3n) = 2, \\ P(1) = P(4) = \cdots &= P(3n-2) = 1, \\ P(2) = P(5) = \cdots &= P(3n-1) = 0, \quad \text{and} \\ &P(3n+1) = 730. \end{aligned}$$

Determine n .

§14 USAMO 1985

1. (*USAMO 1985*) Determine whether or not there are any positive integral solutions of the simultaneous equations

$$x_1^2 + x_2^2 + \cdots + x_{1985}^2 = y^3, \quad x_1^3 + x_2^3 + \cdots + x_{1985}^3 = z^2$$

with distinct integers $x_1, x_2, \dots, x_{1985}$.

2. (*USAMO 1985*) Determine each real root of $x^4 - (2 \cdot 10^{10} + 1)x^2 - x + 10^{20} + 10^{10} - 1 = 0$ correct to four decimal places.
3. (*USAMO 1985*) Let A, B, C, D denote four points in space such that at most one of the distances AB, AC, AD, BC, BD, CD is greater than 1. Determine the maximum value of the sum of the six distances.
4. (*USAMO 1985*) There are n people at a party. Prove that there are two people such that, of the remaining $n-2$ people, there are at least $\lfloor n/2 \rfloor - 1$ of them, each of whom knows both or else knows neither of the two. Assume that "know" is a symmetrical relation; $\lfloor x \rfloor$ denotes the greatest integer less than or equal to x .
5. (*USAMO 1985*) Let $a_1, a_2, a_3, \dots$ be a non-decreasing sequence of positive integers. For $m \geq 1$, define $b_m = \min\{n : a_n \geq m\}$, that is, b_m is the minimum value of n such that $a_n \geq m$. If $a_{19} = 85$, determine the maximum value of $a_1 + a_2 + \cdots + a_{19} + b_1 + b_2 + \cdots + b_{85}$.

§15 USAMO 1986

1. (*USAMO 1986*) (a) Does there exist 14 consecutive positive integers each of which is divisible by one or more primes p from the interval $2 \leq p \leq 11$?
(b) Does there exist 21 consecutive positive integers each of which is divisible by one or more primes p from the interval $2 \leq p \leq 13$?
2. (*USAMO 1986*) During a certain lecture, each of five mathematicians fell asleep exactly twice. For each pair of mathematicians, there was some moment when both were asleep simultaneously. Prove that, at some moment, three of them were sleeping simultaneously.

3. (*USAMO 1986*) What is the smallest integer n , greater than one, for which the root-mean-square of the first n positive integers is an integer?

Note. The root-mean-square of n numbers $a_1, a_2, \dots, a_n$ is defined to be

$$\left[\frac{a_1^2 + a_2^2 + \dots + a_n^2}{n} \right]^{1/2}$$

4. (*USAMO 1986*) Two distinct circles K_1 and K_2 are drawn in the plane. They intersect at points A and B , where AB is the diameter of K_1 . A point P on K_2 and inside K_1 is also given.

Using only a "T-square" (i.e. an instrument which can produce a straight line joining two points and the perpendicular to a line through a point on or off the line), find a construction for two points C and D on K_1 such that CD is perpendicular to AB and $\angle CPD$ is a right angle.

5. (*USAMO 1986*) By a partition π of an integer $n \geq 1$, we mean here a representation of n as a sum of one or more positive integers where the summands must be put in nondecreasing order. (E.g., if $n = 4$, then the partitions π are $1 + 1 + 1 + 1$, $1 + 1 + 2$, $1 + 3$, $2 + 2$, and 4).

For any partition π , define $A(\pi)$ to be the number of 1's which appear in π , and define $B(\pi)$ to be the number of distinct integers which appear in π . (E.g., if $n = 13$ and π is the partition $1 + 1 + 2 + 2 + 2 + 5$, then $A(\pi) = 2$ and $B(\pi) = 3$).

Prove that, for any fixed n , the sum of $A(\pi)$ over all partitions of π of n is equal to the sum of $B(\pi)$ over all partitions of π of n .

§16 USAMO 1987

- (*USAMO 1987*) Find all solutions to $(m^2 + n)(m + n^2) = (m - n)^3$, where m and n are non-zero integers.
- (*USAMO 1987*) The feet of the angle bisectors of $\triangle ABC$ form a right-angled triangle. If the right-angle is at X , where AX is the bisector of $\angle A$, find all possible values for $\angle A$.
- (*USAMO 1987*) X is the smallest set of polynomials $p(x)$ such that:
 - $p = x$ belongs to X .
 - If r belongs to X , then $x \cdot r$ and $(x + (1 - x) \cdot r)$ both belong to X .

Show that if $r(x)$ and $s(x)$ are distinct elements of X , then $r(x) \neq s(x)$ for any $0 < x < 1$.

- (*USAMO 1987*) Three circles C_i are given in the plane: C_1 has diameter AB of length 1; C_2 is concentric and has diameter k ($1 < k < 3$); C_3 has center A and diameter $2k$. We regard k as fixed. Now consider all straight line segments XY which have one endpoint X on C_2 , one endpoint Y on C_3 , and contain the point B . For what ratio $\frac{XB}{BY}$ will the segment XY have minimal length?
- (*USAMO 1987*) $a_1, a_2, \dots, a_n$ is a sequence of 0's and 1's. T is the number of triples (a_i, a_j, a_k) with $i < j < k$ which are not equal to $(0, 1, 0)$ or $(1, 0, 1)$. For

$1 \leq i \leq n$, $f(i)$ is the number of $j < i$ with $a_j = a_i$ plus the number of $j > i$ with $a_j \neq a_i$. Show that $T = \sum_{i=1}^n f(i) \cdot \left(\frac{f(i)-1}{2}\right)$. If n is odd, what is the smallest value of T ?

§17 USAMO 1988

- (USAMO 1988) The repeating decimal $0.ab \cdots k\overline{pq} \cdots u = \frac{m}{n}$, where m and n are relatively prime integers, and there is at least one decimal before the repeating part. Show that n is divisible by 2 or 5 (or both). (For example, $0.011\overline{36} = 0.011363636 \cdots = \frac{1}{88}$, and 88 is divisible by 2.)
- (USAMO 1988) The cubic polynomial $x^3 + ax^2 + bx + c$ has real coefficients and three real roots $r \geq s \geq t$. Show that $k = a^2 - 3b \geq 0$ and that $\sqrt{k} \leq r - t$.
- (USAMO 1988) Let X be the set $\{1, 2, \dots, 20\}$ and let P be the set of all 9-element subsets of X . Show that for any map $f : P \mapsto X$ we can find a 10-element subset Y of X , such that $f(Y - \{k\}) \neq k$ for any k in Y .
- (USAMO 1988) $\triangle ABC$ is a triangle with incenter I . Show that the circumcenters of $\triangle IAB$, $\triangle IBC$, and $\triangle ICA$ lie on a circle whose center is the circumcenter of $\triangle ABC$.
- (USAMO 1988) Let $p(x)$ be the polynomial $(1-x)^a(1-x^2)^b(1-x^3)^c \cdots (1-x^{32})^k$, where $a, b, \dots, k$ are integers. When expanded in powers of x , the coefficient of x^1 is -2 and the coefficients of $x^2, x^3, \dots, x^{32}$ are all zero. Find k .

§18 USAMO 1989

- (USAMO 1989) For each positive integer n , let

$$S_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

$$T_n = S_1 + S_2 + S_3 + \cdots + S_n$$

$$U_n = \frac{T_1}{2} + \frac{T_2}{3} + \frac{T_3}{4} + \cdots + \frac{T_n}{n+1}.$$

 Find, with proof, integers $0 < a, b, c, d < 1000000$ such that $T_{1988} = aS_{1989} - b$ and $U_{1988} = cS_{1989} - d$.
- (USAMO 1989) The 20 members of a local tennis club have scheduled exactly 14 two-person games among themselves, with each member playing in at least one game. Prove that within this schedule there must be a set of 6 games with 12 distinct players.
- (USAMO 1989) Let $P(z) = z^n + c_1 z^{n-1} + c_2 z^{n-2} + \cdots + c_n$ be a polynomial in the complex variable z , with real coefficients c_k . Suppose that $|P(i)| < 1$. Prove that there exist real numbers a and b such that $P(a+bi) = 0$ and $(a^2 + b^2 + 1)^2 < 4b^2 + 1$.
- (USAMO 1989) Let ABC be an acute-angled triangle whose side lengths satisfy the inequalities $AB < AC < BC$. If point I is the center of the inscribed circle of triangle ABC and point O is the center of the circumscribed circle, prove that line IO intersects segments AB and BC .
- (USAMO 1989) Let u and v be real numbers such that

$$(u + u^2 + u^3 + \cdots + u^8) + 10u^9 = (v + v^2 + v^3 + \cdots + v^{10}) + 10v^{11} = 8.$$
 Determine, with proof, which of the two numbers, u or v , is larger.

§19 USAMO 1990

1. (*USAMO 1990*) A certain state issues license plates consisting of six digits (from 0 through 9). The state requires that any two plates differ in at least two places. (Thus the plates $\boxed{027592}$ and $\boxed{020592}$ cannot both be used.) Determine, with proof, the maximum number of distinct license plates that the state can use.
2. (*USAMO 1990*) A sequence of functions $\{f_n(x)\}$ is defined recursively as follows:

$$f_1(x) = \sqrt{x^2 + 48}, \quad \text{and} \\ f_{n+1}(x) = \sqrt{x^2 + 6f_n(x)} \quad \text{for } n \geq 1.$$

(Recall that $\sqrt{}$ is understood to represent the positive square root.) For each positive integer n , find all real solutions of the equation $f_n(x) = 2x$.

3. (*USAMO 1990*) Suppose that necklace A has 14 beads and necklace B has 19. Prove that for any odd integer $n \geq 1$, there is a way to number each of the 33 beads with an integer from the sequence

$$\{n, n+1, n+2, \dots, n+32\}$$

so that each integer is used once, and adjacent beads correspond to relatively prime integers. (Here a "necklace" is viewed as a circle in which each bead is adjacent to two other beads.)

4. (*USAMO 1990*) Find, with proof, the number of positive integers whose base- n representation consists of distinct digits with the property that, except for the leftmost digit, every digit differs by ± 1 from some digit further to the left. (Your answer should be an explicit function of n in simplest form.)
5. (*USAMO 1990*) An acute-angled triangle ABC is given in the plane. The circle with diameter AB intersects altitude CC' and its extension at points M and N , and the circle with diameter AC intersects altitude BB' and its extensions at P and Q . Prove that the points M, N, P, Q lie on a common circle.

§20 USAMO 1991

1. (*USAMO 1991*) In triangle ABC , angle A is twice angle B , angle C is obtuse, and the three side lengths a, b, c are integers. Determine, with proof, the minimum possible perimeter.
2. (*USAMO 1991*) For any nonempty set S of numbers, let $\sigma(S)$ and $\pi(S)$ denote the sum and product, respectively, of the elements of S . Prove that

$$\sum \frac{\sigma(S)}{\pi(S)} = (n^2 + 2n) - \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}\right)(n+1),$$

where " Σ " denotes a sum involving all nonempty subsets S of $\{1, 2, 3, \dots, n\}$.

3. (USAMO 1991) Show that, for any fixed integer $n \geq 1$, the sequence

$$2, 2^2, 2^{2^2}, 2^{2^{2^2}}, \dots \pmod{n}$$

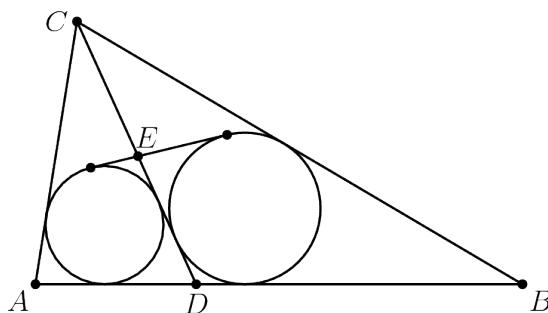
is eventually constant.

[The tower of exponents is defined by $a_1 = 2$, $a_{i+1} = 2^{a_i}$. Also $a_i \pmod{n}$ means the remainder which results from dividing a_i by n .]

4. (USAMO 1991) Let $a = (m^{m+1} + n^{n+1})/(m^m + n^n)$, where m and n are positive integers. Prove that $a^m + a^n \geq m^m + n^n$.

[You may wish to analyze the ratio $(a^N - N^N)/(a - N)$, for real $a \geq 0$ and integer $N \geq 1$.]

5. (USAMO 1991) Let D be an arbitrary point on side AB of a given triangle ABC , and let E be the interior point where CD intersects the external common tangent to the incircles of triangles ACD and BCD . As D assumes all positions between A and B , prove that the point E traces the arc of a circle.



§21 USAMO 1992

1. (USAMO 1992) Find, as a function of n , the sum of the digits of

$$9 \times 99 \times 9999 \times \cdots \times (10^{2^n} - 1),$$

where each factor has twice as many digits as the previous one.

2. (USAMO 1992) Prove

$$\frac{1}{\cos 0^\circ \cos 1^\circ} + \frac{1}{\cos 1^\circ \cos 2^\circ} + \cdots + \frac{1}{\cos 88^\circ \cos 89^\circ} = \frac{\cos 1^\circ}{\sin^2 1^\circ}.$$

3. (USAMO 1992) For a nonempty set S of integers, let $\sigma(S)$ be the sum of the elements of S . Suppose that $A = \{a_1, a_2, \dots, a_{11}\}$ is a set of positive integers with $a_1 < a_2 < \cdots < a_{11}$ and that, for each positive integer $n \leq 1500$, there is a subset S of A for which $\sigma(S) = n$. What is the smallest possible value of a_{10} ?
4. (USAMO 1992) Chords AA' , BB' , and CC' of a sphere meet at an interior point P but are not contained in the same plane. The sphere through A, B, C , and P is tangent to the sphere through A', B', C' , and P . Prove that $AA' = BB' = CC'$.

5. (*USAMO 1992*) Let $P(z)$ be a polynomial with complex coefficients which is of degree 1992 and has distinct zeros. Prove that there exists complex numbers $a_1, a_2, \dots, a_{1992}$ such that $P(z)$ divides the polynomial

$$\left(\cdots \left((z - a_1)^2 - a_2 \right)^2 \cdots - a_{1991} \right)^2 - a_{1992}$$

§22 USAMO 1993

1. (*USAMO 1993*) For each integer $n \geq 2$, determine, with proof, which of the two positive real numbers a and b satisfying

$$a^n = a + 1, \quad b^{2n} = b + 3a$$

is larger.

2. (*USAMO 1993*) Let $ABCD$ be a convex quadrilateral such that diagonals AC and BD intersect at right angles, and let E be their intersection. Prove that the reflections of E across AB, BC, CD, DA are concyclic.

3. (*USAMO 1993*) Consider functions $f : [0, 1] \rightarrow \mathbb{R}$ which satisfy

	(i)	$f(x) \geq 0$ for all x in $[0, 1]$,
	(ii)	$f(1) = 1$,
	(iii)	$f(x) + f(y) \leq f(x + y)$ whenever x, y , and $x + y$ are all in $[0, 1]$.

Find, with proof, the smallest constant c such that

$$f(x) \leq cx$$

for every function f satisfying (i)-(iii) and every x in $[0, 1]$.

4. (*USAMO 1993*) Let a, b be odd positive integers. Define the sequence (f_n) by putting $f_1 = a, f_2 = b$, and by letting f_n for $n \geq 3$ be the greatest odd divisor of $f_{n-1} + f_{n-2}$. Show that f_n is constant for n sufficiently large and determine the eventual value as a function of a and b .

5. (*USAMO 1993*) Let $a_0, a_1, a_2, \dots$ be a sequence of positive real numbers satisfying $a_{i-1}a_{i+1} \leq a_i^2$ for $i = 1, 2, 3, \dots$. (Such a sequence is said to be *log concave*.) Show that for each $n > 1$,

$$\frac{a_0 + \cdots + a_n}{n+1} \cdot \frac{a_1 + \cdots + a_{n-1}}{n-1} \geq \frac{a_0 + \cdots + a_{n-1}}{n} \cdot \frac{a_1 + \cdots + a_n}{n}.$$

§23 USAMO 1994

1. (*USAMO 1994*) Let $k_1 < k_2 < k_3 < \dots$ be positive integers, no two consecutive, and let $s_m = k_1 + k_2 + \dots + k_m$ for $m = 1, 2, 3, \dots$. Prove that, for each positive integer n , the interval $[s_n, s_{n+1})$ contains at least one perfect square.

2. (*USAMO 1994*) The sides of a 99-gon are initially colored so that consecutive sides are red, blue, red, blue, ... red, blue, yellow. We make a sequence of modifications in the coloring, changing the color of one side at a time to one of the three given colors (red, blue, yellow), under the constraint that no two adjacent sides may be the same color. By making a sequence of such modifications, is it possible to arrive at the coloring in which consecutive sides are red, blue, red, blue, red, blue, ... red, yellow, blue?

3. (USAMO 1994) A convex hexagon $ABCDEF$ is inscribed in a circle such that $AB = CD = EF$ and diagonals AD , BE , and CF are concurrent. Let P be the intersection of AD and CE . Prove that $CP/PE = (AC/CE)^2$.
4. (USAMO 1994) Let $a_1, a_2, a_3, \dots$ be a sequence of positive real numbers satisfying $\sum_{j=1}^n a_j \geq \sqrt{n}$ for all $n \geq 1$. Prove that, for all $n \geq 1$,

$$\sum_{j=1}^n a_j^2 > \frac{1}{4} \left(1 + \frac{1}{2} + \dots + \frac{1}{n} \right).$$

5. (USAMO 1994) Let $|U|$, $\sigma(U)$ and $\pi(U)$ denote the number of elements, the sum, and the product, respectively, of a finite set U of positive integers. (If U is the empty set, $|U| = 0$, $\sigma(U) = 0$, $\pi(U) = 1$.) Let S be a finite set of positive integers. As usual, let $\binom{n}{k}$ denote $\frac{n!}{k!(n-k)!}$. Prove that

$$\sum_{U \subseteq S} (-1)^{|U|} \binom{m - \sigma(U)}{|S|} = \pi(S)$$

for all integers $m \geq \sigma(S)$.

§24 USAMO 1995

1. (USAMO 1995) Let p be an odd **prime**. The sequence $(a_n)_{n \geq 0}$ is defined as follows: $a_0 = 0$, $a_1 = 1$, $\dots$, $a_{p-2} = p-2$ and, for all $n \geq p-1$, a_n is the least positive integer that does not form an arithmetic sequence of length p with any of the preceding terms. Prove that, for all n , a_n is the number obtained by writing n in base $p-1$ and reading the result in base p .
2. (USAMO 1995) A calculator is broken so that the only keys that still work are the $\sin$, $\cos$, $\tan$, $\sin^{-1}$, $\cos^{-1}$, and $\tan^{-1}$ buttons. The display initially shows 0. Given any positive rational number q , show that pressing some finite sequence of buttons will yield q . Assume that the calculator does real number calculations with infinite precision. All functions are in terms of radians.
3. (USAMO 1995) Given a nonisosceles, nonright triangle ABC , let O denote the center of its circumscribed circle, and let A_1 , B_1 , and C_1 be the midpoints of sides BC , CA , and AB , respectively. Point A_2 is located on the ray OA_1 so that $\triangle OAA_1$ is similar to $\triangle OA_2A$. Points B_2 and C_2 on rays OB_1 and OC_1 , respectively, are defined similarly. Prove that lines AA_2 , BB_2 , and CC_2 are concurrent, i.e. these three lines intersect at a point.
4. (USAMO 1995) Suppose $q_0, q_1, q_2, \dots$ is an infinite sequence of integers satisfying the following two conditions:
- (i) $m - n$ divides $q_m - q_n$ for $m > n \geq 0$,
 - (ii) there is a polynomial P such that $|q_n| < P(n)$ for all n .
- Prove that there is a polynomial Q such that $q_n = Q(n)$ for all n .
5. (USAMO 1995) Suppose that in a certain society, each pair of persons can be classified as either *amicable* or *hostile*. We shall say that each member of an amicable pair is a *friend* of the other, and each member of a hostile pair is a *foe* of

the other. Suppose that the society has n persons and q amicable pairs, and that for every set of three persons, at least one pair is hostile. Prove that there is at least one member of the society whose foes include $q(1 - 4q/n^2)$ or fewer amicable pairs.

§25 USAMO 1996

1. (USAMO 1996) Prove that the average of the numbers $n \sin n^\circ$ ($n = 2, 4, 6, \dots, 180$) is $\cot 1^\circ$.
2. (USAMO 1996) For any nonempty set S of real numbers, let $\sigma(S)$ denote the sum of the elements of S . Given a set A of n positive integers, consider the collection of all distinct sums $\sigma(S)$ as S ranges over the nonempty subsets of A . Prove that this collection of sums can be partitioned into n classes so that in each class, the ratio of the largest sum to the smallest sum does not exceed 2.
3. (USAMO 1996) Let ABC be a triangle. Prove that there is a line ℓ (in the plane of triangle ABC) such that the intersection of the interior of triangle ABC and the interior of its reflection $A'B'C'$ in ℓ has area more than $\frac{2}{3}$ the area of triangle ABC .
4. (USAMO 1996) An n -term sequence $(x_1, x_2, \dots, x_n)$ in which each term is either 0 or 1 is called a *binary sequence of length n* . Let a_n be the number of binary sequences of length n containing no three consecutive terms equal to 0, 1, 0 in that order. Let b_n be the number of binary sequences of length n that contain no four consecutive terms equal to 0, 0, 1, 1 or 1, 1, 0, 0 in that order. Prove that $b_{n+1} = 2a_n$ for all positive integers n .
5. (USAMO 1996) Let ABC be a triangle, and M an interior point such that $\angle MAB = 10^\circ$, $\angle MBA = 20^\circ$, $\angle MAC = 40^\circ$ and $\angle MCA = 30^\circ$. Prove that the triangle is isosceles.
6. (USAMO 1996) Determine (with proof) whether there is a subset X of the integers with the following property: for any integer n there is exactly one solution of $a + 2b = n$ with $a, b \in X$.

§26 USAMO 1997

1. (USAMO 1997) Let $p_1, p_2, p_3, \dots$ be the prime numbers listed in increasing order, and let x_0 be a real number between 0 and 1. For positive integer k , define

$$x_k = \begin{cases} 0 & \text{if } x_{k-1} = 0 \\ \left\{ \frac{p_k}{x_{k-1}} \right\} & \text{if } x_{k-1} \neq 0 \end{cases}$$

where $\{x\}$ denotes the fractional part of x . (The fractional part of x is given by $x - [x]$ where $[x]$ is the greatest integer less than or equal to x .) Find, with proof, all x_0 satisfying $0 < x_0 < 1$ for which the sequence $x_0, x_1, x_2, \dots$ eventually becomes 0.

2. (USAMO 1997) Let ABC be a triangle, and draw isosceles triangles BCD, CAE, ABF externally to ABC , with BC, CA, AB as their respective bases. Prove that the lines through A, B, C perpendicular to the lines $\overleftrightarrow{EF}, \overleftrightarrow{FD}, \overleftrightarrow{DE}$, respectively, are concurrent.

3. (USAMO 1997) Prove that for any integer n , there exists a unique polynomial Q with coefficients in $\{0, 1, \dots, 9\}$ such that $Q(-2) = Q(-5) = n$.
4. (USAMO 1997) To *clip* a convex n -gon means to choose a pair of consecutive sides AB, BC and to replace them by three segments AM, MN , and NC , where M is the midpoint of AB and N is the midpoint of BC . In other words, one cuts off the triangle MBN to obtain a convex $(n+1)$ -gon. A regular hexagon P_6 of area 1 is clipped to obtain a heptagon P_7 . Then P_7 is clipped (in one of the seven possible ways) to obtain an octagon P_8 , and so on. Prove that no matter how the clippings are done, the area of P_n is greater than $\frac{1}{3}$, for all $n \geq 6$.
5. (USAMO 1997) Prove that, for all positive real numbers a, b, c ,
- $$\frac{1}{a^3+b^3+abc} + \frac{1}{b^3+c^3+abc} + \frac{1}{a^3+c^3+abc} \leq \frac{1}{abc}.$$
6. (USAMO 1997) Suppose the sequence of nonnegative integers $a_1, a_2, \dots, a_{1997}$ satisfies
- $$a_i + a_j \leq a_{i+j} \leq a_i + a_j + 1$$
- for all $i, j \geq 1$ with $i + j \leq 1997$. Show that there exists a real number x such that $a_n = \lfloor nx \rfloor$ (the greatest integer $\leq x$) for all $1 \leq n \leq 1997$.

§27 USAMO 1998

1. (USAMO 1998) Suppose that the set $\{1, 2, \dots, 1998\}$ has been partitioned into disjoint pairs $\{a_i, b_i\}$ ($1 \leq i \leq 999$) so that for all i , $|a_i - b_i|$ equals 1 or 6. Prove that the sum

$$|a_1 - b_1| + |a_2 - b_2| + \dots + |a_{999} - b_{999}|$$

ends in the digit 9.

2. (USAMO 1998) Let $\mathcal{C}_1$ and $\mathcal{C}_2$ be concentric circles, with $\mathcal{C}_2$ in the interior of $\mathcal{C}_1$. From a point A on $\mathcal{C}_1$ one draws the tangent AB to $\mathcal{C}_2$ ($B \in \mathcal{C}_2$). Let C be the second point of intersection of AB and $\mathcal{C}_1$, and let D be the midpoint of AB . A line passing through A intersects $\mathcal{C}_2$ at E and F in such a way that the perpendicular bisectors of DE and CF intersect at a point M on AB . Find, with proof, the ratio AM/MC .
3. (USAMO 1998) Let $a_0, a_1, \dots, a_n$ be numbers from the interval $(0, \pi/2)$ such that

$$\tan\left(a_0 - \frac{\pi}{4}\right) + \tan\left(a_1 - \frac{\pi}{4}\right) + \dots + \tan\left(a_n - \frac{\pi}{4}\right) \geq n - 1.$$

Prove that

$$\tan a_0 \tan a_1 \dots \tan a_n \geq n^{n+1}.$$

4. (USAMO 1998) A computer screen shows a 98×98 chessboard, colored in the usual way. One can select with a mouse any rectangle with sides on the lines of the chessboard and click the mouse button: as a result, the colors in the selected rectangle switch (black becomes white, white becomes black). Find, with proof, the minimum number of mouse clicks needed to make the chessboard all one color.

5. (*USAMO 1998*) Prove that for each $n \geq 2$, there is a set S of n integers such that $(a - b)^2$ divides ab for every distinct $a, b \in S$.
6. (*USAMO 1998*) Let $n \geq 5$ be an integer. Find the largest integer k (as a function of n) such that there exists a convex n -gon $A_1A_2 \dots A_n$ for which exactly k of the quadrilaterals $A_iA_{i+1}A_{i+2}A_{i+3}$ have an inscribed circle. (Here $A_{n+j} = A_j$.)

§28 USAMO 1999

1. (*USAMO 1999*) Some checkers placed on an $n \times n$ checkerboard satisfy the following conditions:
 - (a) every square that does not contain a checker shares a side with one that does;
 - (b) given any pair of squares that contain checkers, there is a sequence of squares containing checkers, starting and ending with the given squares, such that every two consecutive squares of the sequence share a side.

Prove that at least $(n^2 - 2)/3$ checkers have been placed on the board.

2. (*USAMO 1999*) Let $ABCD$ be a cyclic quadrilateral. Prove that

$$|AB - CD| + |AD - BC| \geq 2|AC - BD|.$$

3. (*USAMO 1999*) Let $p > 2$ be a prime and let a, b, c, d be integers not divisible by p , such that

$$\left\{ \frac{ra}{p} \right\} + \left\{ \frac{rb}{p} \right\} + \left\{ \frac{rc}{p} \right\} + \left\{ \frac{rd}{p} \right\} = 2$$

for any integer r not divisible by p . Prove that at least two of the numbers $a + b$, $a + c$, $a + d$, $b + c$, $b + d$, $c + d$ are divisible by p . (Note: $\{x\} = x - \lfloor x \rfloor$ denotes the fractional part of x .)

4. (*USAMO 1999*) Let $a_1, a_2, \dots, a_n$ ($n > 3$) be real numbers such that

$$a_1 + a_2 + \dots + a_n \geq n \quad \text{and} \quad a_1^2 + a_2^2 + \dots + a_n^2 \geq n^2.$$

Prove that $\max(a_1, a_2, \dots, a_n) \geq 2$.

5. (*USAMO 1999*) The Y2K Game is played on a 1×2000 grid as follows. Two players in turn write either an S or an O in an empty square. The first player who produces three consecutive boxes that spell SOS wins. If all boxes are filled without producing SOS then the game is a draw. Prove that the second player has a winning strategy.
6. (*USAMO 1999*) Let $ABCD$ be an isosceles trapezoid with $AB \parallel CD$. The inscribed circle ω of triangle BCD meets CD at E . Let F be a point on the (internal) angle bisector of $\angle DAC$ such that $EF \perp CD$. Let the circumscribed circle of triangle ACF meet line CD at C and G . Prove that the triangle AFG is isosceles.

§29 USAMO 2000

1. (USAMO 2000) Call a real-valued function f very convex if

$$\frac{f(x) + f(y)}{2} \geq f\left(\frac{x+y}{2}\right) + |x-y|$$

holds for all real numbers x and y . Prove that no very convex function exists.

2. (USAMO 2000) Let S be the set of all triangles ABC for which

$$5\left(\frac{1}{AP} + \frac{1}{BQ} + \frac{1}{CR}\right) - \frac{3}{\min\{AP, BQ, CR\}} = \frac{6}{r},$$

where r is the inradius and P, Q, R are the points of tangency of the incircle with sides AB, BC, CA , respectively. Prove that all triangles in S are isosceles and similar to one another.

3. (USAMO 2000) A game of solitaire is played with R red cards, W white cards, and B blue cards. A player plays all the cards one at a time. With each play he accumulates a penalty. If he plays a blue card, then he is charged a penalty which is the number of white cards still in his hand. If he plays a white card, then he is charged a penalty which is twice the number of red cards still in his hand. If he plays a red card, then he is charged a penalty which is three times the number of blue cards still in his hand. Find, as a function of R, W , and B , the minimal total penalty a player can amass and all the ways in which this minimum can be achieved.
4. (USAMO 2000) Find the smallest positive integer n such that if n squares of a 1000×1000 chessboard are colored, then there will exist three colored squares whose centers form a right triangle with legs parallel to the edges of the board.
5. (USAMO 2000) Let $A_1A_2A_3$ be a triangle and let ω_1 be a circle in its plane passing through A_1 and A_2 . Suppose there exist circles $\omega_2, \omega_3, \dots, \omega_7$ such that for $k = 2, 3, \dots, 7$, ω_k is externally tangent to ω_{k-1} and passes through A_k and A_{k+1} , where $A_{n+3} = A_n$ for all $n \geq 1$. Prove that $\omega_7 = \omega_1$.
6. (USAMO 2000) Let $a_1, b_1, a_2, b_2, \dots, a_n, b_n$ be nonnegative real numbers. Prove that

$$\sum_{i,j=1}^n \min\{a_i a_j, b_i b_j\} \leq \sum_{i,j=1}^n \min\{a_i b_j, a_j b_i\}.$$

§30 USAMO 2001

1. (USAMO 2001) Each of eight boxes contains six balls. Each ball has been colored with one of n colors, such that no two balls in the same box are the same color, and no two colors occur together in more than one box. Determine, with justification, the smallest integer n for which this is possible.
2. (USAMO 2001) Let ABC be a triangle and let ω be its incircle. Denote by D_1 and E_1 the points where ω is tangent to sides BC and AC , respectively. Denote by D_2 and E_2 the points on sides BC and AC , respectively, such that $CD_2 = BD_1$ and

- $CE_2 = AE_1$, and denote by P the point of intersection of segments AD_2 and BE_2 . Circle ω intersects segment AD_2 at two points, the closer of which to the vertex A is denoted by Q . Prove that $AQ = D_2P$.
3. (USAMO 2001) Let $a, b, c \geq 0$ and satisfy
 $a^2 + b^2 + c^2 + abc = 4$.
 Show that
 $0 \leq ab + bc + ca - abc \leq 2$.
4. (USAMO 2001) Let P be a point in the plane of triangle ABC such that the segments PA , PB , and PC are the sides of an obtuse triangle. Assume that in this triangle the obtuse angle opposes the side congruent to PA . Prove that $\angle BAC$ is acute.
5. (USAMO 2001) Let S be a set of integers (not necessarily positive) such that
 (a) there exist $a, b \in S$ with $\gcd(a, b) = \gcd(a - 2, b - 2) = 1$;
 (b) if x and y are elements of S (possibly equal), then $x^2 - y$ also belongs to S .
 Prove that S is the set of all integers.
6. (USAMO 2001) Each point in the plane is assigned a real number such that, for any triangle, the number at the center of its inscribed circle is equal to the arithmetic mean of the three numbers at its vertices. Prove that all points in the plane are assigned the same number.

§31 USAMO 2002

1. (USAMO 2002) Let S be a set with 2002 elements, and let N be an integer with $0 \leq N \leq 2^{2002}$. Prove that it is possible to color every subset of S either blue or red so that the following conditions hold:
 (a) the union of any two red subsets is red;
 (b) the union of any two blue subsets is blue;
 (c) there are exactly N red subsets.
2. (USAMO 2002) Let ABC be a triangle such that

$$\left(\cot \frac{A}{2}\right)^2 + \left(2 \cot \frac{B}{2}\right)^2 + \left(3 \cot \frac{C}{2}\right)^2 = \left(\frac{6s}{7r}\right)^2,$$
 where s and r denote its **semiperimeter** and **inradius**, respectively. Prove that triangle ABC is similar to a triangle T whose side lengths are all positive integers with no common divisor and determine those integers.
3. (USAMO 2002) Prove that any monic polynomial (a polynomial with leading coefficient 1) of degree n with real coefficients is the average of two monic polynomials of degree n with n real roots.
4. (USAMO 2002) Let $\mathbb{R}$ be the set of **real numbers**. Determine all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 - y^2) = xf(x) - yf(y)$$
 for all pairs of real numbers x and y .

5. (*USAMO 2002*) Let a, b be integers greater than 2. Prove that there exists a positive integer k and a finite sequence $n_1, n_2, \dots, n_k$ of positive integers such that $n_1 = a$, $n_k = b$, and $n_i n_{i+1}$ is divisible by $n_i + n_{i+1}$ for each i ($1 \leq i < k$).
6. (*USAMO 2002*) I have an $n \times n$ sheet of stamps, from which I've been asked to tear out blocks of three adjacent stamps in a single row or column. (I can only tear along the perforations separating adjacent stamps, and each block must come out of the sheet in one piece.) Let $b(n)$ be the smallest number of blocks I can tear out and make it impossible to tear out any more blocks. Prove that there are real constants c and d such that
- $$\frac{1}{7}n^2 - cn \leq b(n) \leq \frac{1}{5}n^2 + dn$$
- for all $n > 0$.

§32 USAMO 2003

1. (*USAMO 2003*) Prove that for every positive integer n there exists an n -digit number divisible by 5^n all of whose digits are odd.
2. (*USAMO 2003*) A convex polygon $\mathcal{P}$ in the plane is dissected into smaller convex polygons by drawing all of its diagonals. The lengths of all sides and all diagonals of the polygon $\mathcal{P}$ are rational numbers. Prove that the lengths of all sides of all polygons in the dissection are also rational numbers.
3. (*USAMO 2003*) Let $n \neq 0$. For every sequence of integers
- $$A = a_0, a_1, a_2, \dots, a_n$$
- satisfying $0 \leq a_i \leq i$, for $i = 0, \dots, n$, define another sequence
- $$t(A) = t(a_0), t(a_1), t(a_2), \dots, t(a_n)$$
- by setting $t(a_i)$ to be the number of terms in the sequence A that precede the term a_i and are different from a_i . Show that, starting from any sequence A as above, fewer than n applications of the transformation t lead to a sequence B such that $t(B) = B$.
4. (*USAMO 2003*) Let ABC be a triangle. A circle passing through A and B intersects segments AC and BC at D and E , respectively. Lines AB and DE intersect at F , while lines BD and CE intersect at M . Prove that $MF = MC$ if and only if $MB \cdot MD = MC^2$.
5. (*USAMO 2003*) Let a, b, c be positive real numbers. Prove that
- $$\frac{(2a+b+c)^2}{2a^2+(b+c)^2} + \frac{(2b+c+a)^2}{2b^2+(c+a)^2} + \frac{(2c+a+b)^2}{2c^2+(a+b)^2} \leq 8.$$
6. (*USAMO 2003*) At the vertices of a regular hexagon are written six nonnegative integers whose sum is 2003. Bert is allowed to make moves of the following form: he may pick a vertex and replace the number written there by the absolute value of the difference between the numbers written at the two neighboring vertices. Prove that Bert can make a sequence of moves, after which the number 0 appears at all six vertices.

§33 USAMO 2004

1. (*USAMO 2004*) Let $ABCD$ be a **quadrilateral** circumscribed about a circle, whose interior and exterior angles are at least 60° . Prove that

$$\frac{1}{3}|AB^3 - AD^3| \leq |BC^3 - CD^3| \leq 3|AB^3 - AD^3|$$

When does equality hold?

2. (*USAMO 2004*) Suppose $a_1, \dots, a_n$ are integers whose greatest common divisor is 1. Let S be a set of integers with the following properties:

- (a) For $i = 1, \dots, n$, $a_i \in S$.
- (b) For $i, j = 1, \dots, n$ (not necessarily distinct), $a_i - a_j \in S$.
- (c) For any integers $x, y \in S$, if $x + y \in S$, then $x - y \in S$.

Prove that S must be the set of all integers.

3. (*USAMO 2004*) For what values of $k > 0$ is it possible to dissect a $1 \times k$ rectangle into two similar, but incongruent, polygons?

4. (*USAMO 2004*) Alice and Bob play a game on a 6 by 6 grid. On his or her turn, a player chooses a rational number not yet appearing on the grid and writes it in an empty square of the grid. Alice goes first and then the players alternate. When all squares have numbers written in them, in each row, the square with the greatest number is colored black. Alice wins if she can then draw a line from the top of the grid to the bottom of the grid that stays in black squares, and Bob wins if she can't. (If two squares share a vertex, Alice can draw a line from one to the other that stays in those two squares. Find, with proof, a winning strategy for one of the players.

5. (*USAMO 2004*) Let a, b , and c be positive real numbers. Prove that

$$(a^5 - a^2 + 3)(b^5 - b^2 + 3)(c^5 - c^2 + 3) \geq (a + b + c)^3.$$

6. (*USAMO 2004*) A circle ω is inscribed in a quadrilateral $ABCD$. Let I be the center of ω . Suppose that

$$(AI + DI)^2 + (BI + CI)^2 = (AB + CD)^2.$$

Prove that $ABCD$ is an **isosceles trapezoid**.

§34 USAMO 2005

1. (*USAMO 2005*) (*Zuming Feng*) Determine all composite positive integers n for which it is possible to arrange all divisors of n that are greater than 1 in a circle so that no two adjacent divisors are relatively prime.
2. (*USAMO 2005*) (*Răzvan Gelca*) Prove that the system

$$\begin{aligned} x^6 + x^3 + x^3y + y &= 147^{157} \\ x^3 + x^3y + y^2 + y + z^9 &= 157^{147} \end{aligned}$$

has no solutions in integers x, y , and z .

3. (USAMO 2005) (*Zuming Feng*) Let ABC be an acute-angled triangle, and let P and Q be two points on side BC . Construct point C_1 in such a way that convex quadrilateral $APBC_1$ is cyclic, $QC_1 \parallel CA$, and C_1 and Q lie on opposite sides of line AB . Construct point B_1 in such a way that convex quadrilateral $APCB_1$ is cyclic, $QB_1 \parallel BA$, and B_1 and Q lie on opposite sides of line AC . Prove that points B_1, C_1, P , and Q lie on a circle.
4. (USAMO 2005) Legs L_1, L_2, L_3, L_4 of a square table each have length n , where n is a positive integer. For how many ordered 4-tuples (k_1, k_2, k_3, k_4) of nonnegative integers can we cut a piece of length k_i from the end of leg L_i ($i = 1, 2, 3, 4$) and still have a stable table?
- (The table is stable if it can be placed so that all four of the leg ends touch the floor. Note that a cut leg of length 0 is permitted.)
5. (USAMO 2005) Let n be an integer greater than 1. Suppose $2n$ points are given in the plane, no three of which are collinear. Suppose n of the given $2n$ points are colored blue and the other n colored red. A line in the plane is called a *balancing line* if it passes through one blue and one red point and, for each side of the line, the number of blue points on that side is equal to the number of red points on the same side.
- Prove that there exist at least two balancing lines.
6. (USAMO 2005) For m a positive integer, let $s(m)$ be the sum of the digits of m . For $n \geq 2$, let $f(n)$ be the minimal k for which there exists a set S of n positive integers such that $s(\sum_{x \in X} x) = k$ for any nonempty subset $X \subset S$. Prove that there are constants $0 < C_1 < C_2$ with

$$C_1 \log_{10} n \leq f(n) \leq C_2 \log_{10} n.$$

§35 USAMO 2006

1. (USAMO 2006) Let p be a prime number and let s be an integer with $0 < s < p$. Prove that there exist integers m and n with $0 < m < n < p$ and

$$\left\{ \frac{sm}{p} \right\} < \left\{ \frac{sn}{p} \right\} < \frac{s}{p}$$

if and only if s is not a divisor of $p - 1$.

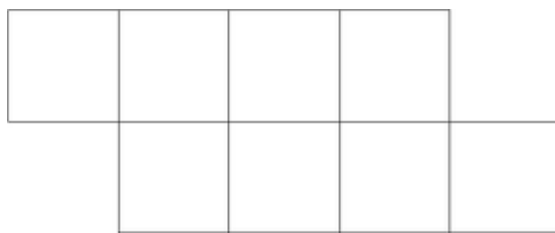
Note: For x a real number, let $\lfloor x \rfloor$ denote the greatest integer less than or equal to x , and let $\{x\} = x - \lfloor x \rfloor$ denote the fractional part of x .

2. (USAMO 2006) For a given positive integer k find, in terms of k , the minimum value of N for which there is a set of $2k + 1$ distinct positive integers that has sum greater than N but every subset of size k has sum at most $N/2$.
3. (USAMO 2006) For integral m , let $p(m)$ be the greatest prime divisor of m . By convention, we set $p(\pm 1) = 1$ and $p(0) = \infty$. Find all polynomials f with integer coefficients such that the sequence $\{p(f(n^2)) - 2n\}_{n \geq 0}$ is bounded above. (In particular, this requires $f(n^2) \neq 0$ for $n \geq 0$.)
4. (USAMO 2006) Find all positive integers n for which there exists an integer $k \geq 2$ and positive rationals $a_1, a_2, \dots, a_k$ satisfying $a_1 + a_2 + \dots + a_k = a_1 \cdot a_2 \cdots a_k = n$.

5. (USAMO 2006) A mathematical frog jumps along the number line. The frog starts at 1, and jumps according to the following rule: if the frog is at integer n , then it can jump either to $n + 1$ or to $n + 2^{m_n+1}$ where 2^{m_n} is the largest power of 2 that is a factor of n . Show that if $k \geq 2$ is a positive integer and i is a nonnegative integer, then the minimum number of jumps needed to reach $2^i k$ is greater than the minimum number of jumps needed to reach 2^i .
6. (USAMO 2006) Let $ABCD$ be a quadrilateral, and let E and F be points on sides AD and BC , respectively, such that $AE/ED = BF/FC$. Ray FE meets rays BA and CD at S and T respectively. Prove that the circumcircles of triangles SAE , SBF , TCF , and TDE pass through a common point.

§36 USAMO 2007

1. (USAMO 2007) Let n be a positive integer. Define a sequence by setting $a_1 = n$ and, for each $k > 1$, letting a_k be the unique integer in the range $0 \leq a_k \leq k - 1$ for which $a_1 + a_2 + \cdots + a_k$ is divisible by k . For instance, when $n = 9$ the obtained sequence is $9, 1, 2, 0, 3, 3, 3, \dots$. Prove that for any n the sequence $a_1, a_2, a_3, \dots$ eventually becomes constant.
2. (USAMO 2007) A square grid on the Euclidean plane consists of all points (m, n) , where m and n are integers. Is it possible to cover all grid points by an infinite family of discs with non-overlapping interiors if each disc in the family has radius at least 5?
3. (USAMO 2007) Let S be a set containing $n^2 + n - 1$ elements, for some positive integer n . Suppose that the n -element subsets of S are partitioned into two classes. Prove that there are at least n pairwise disjoint sets in the same class.
4. (USAMO 2007) An *animal* with n cells is a connected figure consisting of n equal-sized cells.¹ The figure below shows an 8-cell animal.



A *dinosaur* is an animal with at least 2007 cells. It is said to be *primitive* if its cells cannot be partitioned into two or more dinosaurs. Find with proof the maximum number of cells in a primitive dinosaur.

¹Animals are also called *polyominoes*. They can be defined inductively. Two cells are *adjacent* if they share a complete edge. A single cell is an animal, and given an animal with n cells, one with $n + 1$ cells is obtained by adjoining a new cell by making it adjacent to one or more existing cells.

5. (USAMO 2007) Prove that for every nonnegative integer n , the number $7^{7^n} + 1$ is the product of at least $2n + 3$ (not necessarily distinct) primes.

6. (USAMO 2007) Let ABC be an acute triangle with ω , Ω , and R being its incircle, circumcircle, and circumradius, respectively. Circle ω_A is tangent internally to Ω at A and tangent externally to ω . Circle Ω_A is tangent internally to Ω at A and tangent internally to ω . Let P_A and Q_A denote the centers of ω_A and Ω_A , respectively. Define points P_B, Q_B, P_C, Q_C analogously. Prove that

$$8P_AQ_A \cdot P_BQ_B \cdot P_CQ_C \leq R^3,$$

with equality if and only if triangle ABC is equilateral.

§37 USAMO 2008

- (USAMO 2008) (Titu Andreescu) Prove that for each positive integer n , there are pairwise relatively prime integers $k_0, k_1, \dots, k_n$, all strictly greater than 1, such that $k_0k_1 \cdots k_n - 1$ is the product of two consecutive integers.
- (USAMO 2008) (Zuming Feng) Let ABC be an acute, **scalene** triangle, and let M, N , and P be the midpoints of $\overline{BC}$, $\overline{CA}$, and $\overline{AB}$, respectively. Let the **perpendicular bisectors** of $\overline{AB}$ and $\overline{AC}$ intersect ray AM in points D and E respectively, and let lines BD and CE intersect in point F , inside of triangle ABC . Prove that points A, N, F , and P all lie on one circle.
- (USAMO 2008) (Gabriel Carroll) Let n be a positive integer. Denote by S_n the set of points (x, y) with integer coordinates such that

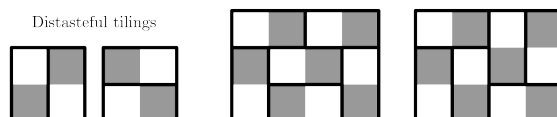
$$|x| + \left| y + \frac{1}{2} \right| < n$$

A path is a sequence of distinct points $(x_1, y_1), (x_2, y_2), \dots, (x_\ell, y_\ell)$ in S_n such that, for $i = 2, \dots, \ell$, the distance between (x_i, y_i) and (x_{i-1}, y_{i-1}) is 1 (in other words, the points (x_i, y_i) and (x_{i-1}, y_{i-1}) are neighbors in the lattice of points with integer coordinates). Prove that the points in S_n cannot be partitioned into fewer than n paths (a partition of S_n into m paths is a set $\mathcal{P}$ of m nonempty paths such that each point in S_n appears in exactly one of the m paths in $\mathcal{P}$).

- (USAMO 2008) (Gregory Galparin) Let $\mathcal{P}$ be a **convex polygon** with n sides, $n \geq 3$. Any set of $n - 3$ diagonals of $\mathcal{P}$ that do not intersect in the interior of the polygon determine a *triangulation* of $\mathcal{P}$ into $n - 2$ triangles. If $\mathcal{P}$ is regular and there is a triangulation of $\mathcal{P}$ consisting of only isosceles triangles, find all the possible values of n .
- (USAMO 2008) (Kiran Kedlaya) Three nonnegative real numbers r_1, r_2, r_3 are written on a blackboard. These numbers have the property that there exist integers a_1, a_2, a_3 , not all zero, satisfying $a_1r_1 + a_2r_2 + a_3r_3 = 0$. We are permitted to perform the following operation: find two numbers x, y on the blackboard with $x \leq y$, then erase y and write $y - x$ in its place. Prove that after a finite number of such operations, we can end up with at least one 0 on the blackboard.
- (USAMO 2008) (**Sam Vandervelde**) At a certain mathematical conference, every pair of mathematicians are either friends or strangers. At mealtime, every participant eats in one of two large dining rooms. Each mathematician insists upon eating in a room which contains an even number of his or her friends. Prove that the number of ways that the mathematicians may be split between the two rooms is a power of two (i.e., is of the form 2^k for some positive integer k).

§38 USAMO 2009

- (USAMO 2009) Given circles ω_1 and ω_2 intersecting at points X and Y , let ℓ_1 be a line through the center of ω_1 intersecting ω_2 at points P and Q and let ℓ_2 be a line through the center of ω_2 intersecting ω_1 at points R and S . Prove that if P, Q, R and S lie on a circle then the center of this circle lies on line XY .
- (USAMO 2009) Let n be a positive integer. Determine the size of the largest subset of $\{-n, -n+1, \dots, n-1, n\}$ which does not contain three elements a, b, c (not necessarily distinct) satisfying $a + b + c = 0$.
- (USAMO 2009) We define a *chessboard polygon* to be a polygon whose sides are situated along lines of the form $x = a$ or $y = b$, where a and b are integers. These lines divide the interior into unit squares, which are shaded alternately grey and white so that adjacent squares have different colors. To tile a chessboard polygon by dominoes is to exactly cover the polygon by non-overlapping 1×2 rectangles. Finally, a *tasteful tiling* is one which avoids the two configurations of dominoes shown on the left below. Two tilings of a 3×4 rectangle are shown; the first one is tasteful, while the second is not, due to the vertical dominoes in the upper right corner.



- Prove that if a chessboard polygon can be tiled by dominoes, then it can be done so tastefully.
 - Prove that such a tasteful tiling is unique.
- (USAMO 2009) For $n \geq 2$ let $a_1, a_2, \dots, a_n$ be positive real numbers such that

$$(a_1 + a_2 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} \right) \leq \left(n + \frac{1}{2} \right)^2$$
 Prove that $\max(a_1, a_2, \dots, a_n) \leq 4 \min(a_1, a_2, \dots, a_n)$.
 - (USAMO 2009) Trapezoid $ABCD$, with $\overline{AB} \parallel \overline{CD}$, is inscribed in circle ω and point G lies inside triangle BCD . Rays AG and BG meet ω again at points P and Q , respectively. Let the line through G parallel to $\overline{AB}$ intersect $\overline{BD}$ and $\overline{BC}$ at points R and S , respectively. Prove that quadrilateral $PQRS$ is cyclic if and only if $\overline{BG}$ bisects $\angle CBD$.
 - (USAMO 2009) Let $s_1, s_2, s_3, \dots$ be an infinite, nonconstant sequence of rational numbers, meaning it is not the case that $s_1 = s_2 = s_3 = \dots$. Suppose that $t_1, t_2, t_3, \dots$ is also an infinite, nonconstant sequence of rational numbers with the property that $(s_i - s_j)(t_i - t_j)$ is an integer for all i and j . Prove that there exists a rational number r such that $(s_i - s_j)r$ and $(t_i - t_j)/r$ are integers for all i and j .

§39 USAMO 2010

- (USAMO 2010) Let $AXYZB$ be a convex pentagon inscribed in a semicircle of diameter AB . Denote by P, Q, R, S the feet of the perpendiculars from Y onto lines AX, BX, AZ, BZ , respectively. Prove that the acute angle formed by lines PQ and RS is half the size of $\angle XOZ$, where O is the midpoint of segment AB .

2. (*USAMO 2010*) There are n students standing in a circle, one behind the other. The students have heights $h_1 < h_2 < \cdots < h_n$. If a student with height h_k is standing directly behind a student with height h_{k-2} or less, the two students are permitted to switch places. Prove that it is not possible to make more than $\binom{n}{3}$ such switches before reaching a position in which no further switches are possible.
3. (*USAMO 2010*) The 2010 positive numbers $a_1, a_2, \dots, a_{2010}$ satisfy the inequality $a_i a_j \leq i + j$ for all distinct indices i, j . Determine, with proof, the largest possible value of the product $a_1 a_2 \cdots a_{2010}$.
4. (*USAMO 2010*) Let ABC be a triangle with $\angle A = 90^\circ$. Points D and E lie on sides AC and AB , respectively, such that $\angle ABD = \angle DBC$ and $\angle ACE = \angle ECB$. Segments BD and CE meet at I . Determine whether or not it is possible for segments AB, AC, BI, ID, CI, IE to all have integer side lengths.
5. (*USAMO 2010*) Let $q = \frac{3p-5}{2}$ where p is an odd prime, and let

$$S_q = \frac{1}{2 \cdot 3 \cdot 4} + \frac{1}{5 \cdot 6 \cdot 7} + \cdots + \frac{1}{q \cdot (q+1) \cdot (q+2)}.$$

Prove that if $\frac{1}{p} - 2S_q = \frac{m}{n}$ for relatively prime integers m and n , then $m - n$ is divisible by p .

6. (*USAMO 2010*) A blackboard contains 68 pairs of nonzero integers. Suppose that for each positive integer k at most one of the pairs (k, k) and $(-k, -k)$ is written on the blackboard. A student erases some of the 136 integers, subject to the condition that no two erased integers may add to 0. The student then scores one point for each of the 68 pairs in which at least one integer is erased. Determine, with proof, the largest number N of points that the student can guarantee to score regardless of which 68 pairs have been written on the board.

§40 USAMO 2011

1. (*USAMO 2011*) Let a, b, c be positive real numbers such that $a^2 + b^2 + c^2 + (a + b + c)^2 \leq 4$. Prove that

$$\frac{ab+1}{(a+b)^2} + \frac{bc+1}{(b+c)^2} + \frac{ca+1}{(c+a)^2} \geq 3.$$

2. (*USAMO 2011*) An integer is assigned to each vertex of a regular pentagon so that the sum of the five integers is 2011. A turn of a solitaire game consists of subtracting an integer m from each of the integers at two neighboring vertices and adding $2m$ to the opposite vertex, which is not adjacent to either of the first two vertices. (The amount m and the vertices chosen can vary from turn to turn.) The game is won at a certain vertex if, after some number of turns, that vertex has the number 2011 and the other four vertices have the number 0. Prove that for any choice of the initial integers, there is exactly one vertex at which the game can be won.
3. (*USAMO 2011*) In hexagon $ABCDEF$, which is nonconvex but not self-intersecting, no pair of opposite sides are parallel. The internal angles satisfy $\angle A = 3\angle D$,

- $\angle C = 3\angle F$, and $\angle E = 3\angle B$. Furthermore, $AB = DE$, $BC = EF$, and $CD = FA$. Prove that diagonals $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$ are concurrent.
4. (USAMO 2011) Consider the assertion that for each positive integer $n \geq 2$, the remainder upon dividing 2^{2^n} by $2^n - 1$ is a power of 4. Either prove the assertion or find (with proof) a counterexample.
5. (USAMO 2011) Let P be a given point inside quadrilateral $ABCD$. Points Q_1 and Q_2 are located within $ABCD$ such that $\angle Q_1BC = \angle ABP$, $\angle Q_1CB = \angle DCP$, $\angle Q_2AD = \angle BAP$, $\angle Q_2DA = \angle CDP$. Prove that $\overline{Q_1Q_2} \parallel \overline{BA}$ if and only if $\overline{Q_1Q_2} \parallel \overline{CD}$.
6. (USAMO 2011) Let A be a set with $|A| = 225$, meaning that A has 225 elements. Suppose further that there are eleven subsets $A_1, \dots, A_{11}$ of A such that $|A_i| = 45$ for $1 \leq i \leq 11$ and $|A_i \cap A_j| = 9$ for $1 \leq i < j \leq 11$. Prove that $|A_1 \cup A_2 \cup \dots \cup A_{11}| \geq 165$, and give an example for which equality holds.

§41 USAMO 2012

1. (USAMO 2012) Find all integers $n \geq 3$ such that among any n positive real numbers $a_1, a_2, \dots, a_n$ with

$$\max(a_1, a_2, \dots, a_n) \leq n \cdot \min(a_1, a_2, \dots, a_n),$$

there exist three that are the side lengths of an acute triangle.

2. (USAMO 2012) A circle is divided into 432 congruent arcs by 432 points. The points are colored in four colors such that 108 points are colored red, 108 points are colored green, 108 points are colored blue and the remaining 108 points are colored yellow. Prove that one can choose three points of each color in such a way that the four triangles formed by the chosen points of the same color are congruent.
3. (USAMO 2012) Determine which integers $n > 1$ have the property that there exists an infinite sequence $a_1, a_2, a_3, \dots$ of nonzero integers such that the equality

$$a_k + 2a_{2k} + \dots + na_{nk} = 0$$

holds for every positive integer k .

4. (USAMO 2012) Find all functions $f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ (where $\mathbb{Z}^+$ is the set of positive integers) such that $f(n!) = f(n)!$ for all positive integers n and such that $m - n$ divides $f(m) - f(n)$ for all distinct positive integers m, n .
5. (USAMO 2012) Let P be a point in the plane of triangle ABC , and γ a line passing through P . Let A', B', C' be the points where the reflections of lines PA, PB, PC with respect to γ intersect lines BC, AC, AB , respectively. Prove that A', B', C' are collinear.
6. (USAMO 2012) For integer $n \geq 2$, let $x_1, x_2, \dots, x_n$ be real numbers satisfying

$$x_1 + x_2 + \dots + x_n = 0, \quad \text{and} \quad x_1^2 + x_2^2 + \dots + x_n^2 = 1.$$

For each subset $A \subseteq \{1, 2, \dots, n\}$, define

$$S_A = \sum_{i \in A} x_i.$$

(If A is the empty set, then $S_A = 0$.)

Prove that for any positive number λ , the number of sets A satisfying $S_A \geq \lambda$ is at most $2^{n-3}/\lambda^2$. For what choices of $x_1, x_2, \dots, x_n$, λ does equality hold?

§42 USAMO 2013

- (USAMO 2013) In triangle ABC , points P, Q, R lie on sides BC, CA, AB respectively. Let $\omega_A, \omega_B, \omega_C$ denote the circumcircles of triangles AQR, BRP, CPQ , respectively. Given the fact that segment AP intersects $\omega_A, \omega_B, \omega_C$ again at X, Y, Z respectively, prove that $YX/XZ = BP/PC$.
- (USAMO 2013) For a positive integer $n \geq 3$ plot n equally spaced points around a circle. Label one of them A , and place a marker at A . One may move the marker forward in a clockwise direction to either the next point or the point after that. Hence there are a total of $2n$ distinct moves available; two from each point. Let a_n count the number of ways to advance around the circle exactly twice, beginning and ending at A , without repeating a move. Prove that $a_{n-1} + a_n = 2^n$ for all $n \geq 4$.
- (USAMO 2013) Let n be a positive integer. There are $\frac{n(n+1)}{2}$ marks, each with a black side and a white side, arranged into an equilateral triangle, with the biggest row containing n marks. Initially, each mark has the black side up. An *operation* is to choose a line parallel to the sides of the triangle, and flipping all the marks on that line. A configuration is called *admissible* if it can be obtained from the initial configuration by performing a finite number of operations. For each admissible configuration C , let $f(C)$ denote the smallest number of operations required to obtain C from the initial configuration. Find the maximum value of $f(C)$, where C varies over all admissible configurations.
- (USAMO 2013) Find all real numbers $x, y, z \geq 1$ satisfying

$$\min(\sqrt{x + xyz}, \sqrt{y + xyz}, \sqrt{z + xyz}) = \sqrt{x-1} + \sqrt{y-1} + \sqrt{z-1}.$$

- (USAMO 2013) Given positive integers m and n , prove that there is a positive integer c such that the numbers cm and cn have the same number of occurrences of each non-zero digit when written in base ten.
- (USAMO 2013) Let ABC be a triangle. Find all points P on segment BC satisfying the following property: If X and Y are the intersections of line PA with the common external tangent lines of the circumcircles of triangles PAB and PAC , then

$$\left(\frac{PA}{XY}\right)^2 + \frac{PB \cdot PC}{AB \cdot AC} = 1.$$

§43 USAMO 2014

- (USAMO 2014) Let a, b, c, d be real numbers such that $b - d \geq 5$ and all zeros x_1, x_2, x_3 , and x_4 of the polynomial $P(x) = x^4 + ax^3 + bx^2 + cx + d$ are real. Find

the smallest value the product

$$(x_1^2 + 1)(x_2^2 + 1)(x_3^2 + 1)(x_4^2 + 1)$$

can take.

2. (USAMO 2014) Let $\mathbb{Z}$ be the set of integers. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$xf(2f(y) - x) + y^2f(2x - f(y)) = \frac{(f(x))^2}{x} + f(yf(y))$$

for all $x, y \in \mathbb{Z}$ with $x \neq 0$.

3. (USAMO 2014) Prove that there exists an infinite set of points

$$\dots, P_{-3}, P_{-2}, P_{-1}, P_0, P_1, P_2, P_3, \dots$$

in the plane with the following property: For any three distinct integers a, b , and c , points P_a, P_b , and P_c are collinear if and only if $a + b + c = 2014$.

4. (USAMO 2014) Let k be a positive integer. Two players A and B play a game on an infinite grid of regular hexagons. Initially all the grid cells are empty. Then the players alternately take turns with A moving first. In his move, A may choose two adjacent hexagons in the grid which are empty and place a counter in both of them. In his move, B may choose any counter on the board and remove it. If at any time there are k consecutive grid cells in a line all of which contain a counter, A wins. Find the minimum value of k for which A cannot win in a finite number of moves, or prove that no such minimum value exists.
5. (USAMO 2014) Let ABC be a triangle with orthocenter H and let P be the second intersection of the circumcircle of triangle AHC with the internal bisector of the angle $\angle BAC$. Let X be the circumcenter of triangle APB and Y the orthocenter of triangle APC . Prove that the length of segment XY is equal to the circumradius of triangle ABC .
6. (USAMO 2014) Prove that there is a constant $c > 0$ with the following property: If a, b, n are positive integers such that $\gcd(a + i, b + j) > 1$ for all $i, j \in \{0, 1, \dots, n\}$, then

$$\min\{a, b\} > c^n \cdot n^{\frac{n}{2}}.$$

§44 USAMO 2015

1. (USAMO 2015) Solve in integers the equation

$$x^2 + xy + y^2 = \left(\frac{x+y}{3} + 1\right)^3.$$

2. (USAMO 2015) Quadrilateral $APBQ$ is inscribed in circle ω with $\angle P = \angle Q = 90^\circ$ and $AP = AQ < BP$. Let X be a variable point on segment $\overline{PQ}$. Line AX meets ω again at S (other than A). Point T lies on arc AQB of ω such that $\overline{XT}$ is perpendicular to $\overline{AX}$. Let M denote the midpoint of chord $\overline{ST}$. As X varies on segment $\overline{PQ}$, show that M moves along a circle.

3. (*USAMO 2015*) Let $S = \{1, 2, \dots, n\}$, where $n \geq 1$. Each of the 2^n subsets of S is to be colored red or blue. (The subset itself is assigned a color and not its individual elements.) For any set $T \subseteq S$, we then write $f(T)$ for the number of subsets of T that are blue.

Determine the number of colorings that satisfy the following condition: for any subsets T_1 and T_2 of S ,

$$f(T_1)f(T_2) = f(T_1 \cup T_2)f(T_1 \cap T_2).$$

4. (*USAMO 2015*) Steve is piling $m \geq 1$ indistinguishable stones on the squares of an $n \times n$ grid. Each square can have an arbitrarily high pile of stones. After he finished piling his stones in some manner, he can then perform stone moves, defined as follows. Consider any four grid squares, which are corners of a rectangle, i.e. in positions $(i, k), (i, l), (j, k), (j, l)$ for some $1 \leq i, j, k, l \leq n$, such that $i < j$ and $k < l$. A stone move consists of either removing one stone from each of (i, k) and (j, l) and moving them to (i, l) and (j, k) respectively, or removing one stone from each of (i, l) and (j, k) and moving them to (i, k) and (j, l) respectively.

Two ways of piling the stones are equivalent if they can be obtained from one another by a sequence of stone moves.

How many different non-equivalent ways can Steve pile the stones on the grid?

5. (*USAMO 2015*) Let a, b, c, d, e be distinct positive integers such that $a^4 + b^4 = c^4 + d^4 = e^5$. Show that $ac + bd$ is a composite number.
6. (*USAMO 2015*) Consider $0 < \lambda < 1$, and let A be a multiset of positive integers. Let $A_n = \{a \in A : a \leq n\}$. Assume that for every $n \in \mathbb{N}$, the set A_n contains at most $n\lambda$ numbers. Show that there are infinitely many $n \in \mathbb{N}$ for which the sum of the elements in A_n is at most $\frac{n(n+1)}{2}\lambda$. (A multiset is a set-like collection of elements in which order is ignored, but repetition of elements is allowed and multiplicity of elements is significant. For example, multisets $\{1, 2, 3\}$ and $\{2, 1, 3\}$ are equivalent, but $\{1, 1, 2, 3\}$ and $\{1, 2, 3\}$ differ.)

§45 USAMO 2016

1. (*USAMO 2016*) Let $X_1, X_2, \dots, X_{100}$ be a sequence of mutually distinct nonempty subsets of a set S . Any two sets X_i and X_{i+1} are disjoint and their union is not the whole set S , that is, $X_i \cap X_{i+1} = \emptyset$ and $X_i \cup X_{i+1} \neq S$, for all $i \in \{1, \dots, 99\}$. Find the smallest possible number of elements in S .
2. (*USAMO 2016*) Prove that for any positive integer k ,

$$(k^2)! \cdot \prod_{j=0}^{k-1} \frac{j!}{(j+k)!}$$

is an integer.

3. (*USAMO 2016*) Let $\triangle ABC$ be an acute triangle, and let I_B, I_C , and O denote its B -excenter, C -excenter, and circumcenter, respectively. Points E and Y are selected on $\overline{AC}$ such that $\angle ABY = \angle CBY$ and $\overline{BE} \perp \overline{AC}$. Similarly, points F and Z are selected on $\overline{AB}$ such that $\angle ACZ = \angle BCZ$ and $\overline{CF} \perp \overline{AB}$. Lines $\overleftrightarrow{I_B F}$ and $\overleftrightarrow{I_C E}$ meet at P . Prove that $\overline{PO}$ and $\overline{YZ}$ are perpendicular.

4. (USAMO 2016) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that for all real numbers x and y ,

$$(f(x) + xy) \cdot f(x - 3y) + (f(y) + xy) \cdot f(3x - y) = (f(x + y))^2.$$

5. (USAMO 2016) An equilateral pentagon $AMNPQ$ is inscribed in triangle ABC such that $M \in \overline{AB}$, $Q \in \overline{AC}$, and $N, P \in \overline{BC}$. Let S be the intersection of $\overleftrightarrow{MN}$ and $\overleftrightarrow{PQ}$. Denote by ℓ the angle bisector of $\angle MSQ$.

Prove that $\overline{OI}$ is parallel to ℓ , where O is the circumcenter of triangle ABC , and I is the incenter of triangle ABC .

6. (USAMO 2016) Integers n and k are given, with $n \geq k \geq 2$. You play the following game against an evil wizard.

The wizard has $2n$ cards; for each $i = 1, \dots, n$, there are two cards labeled i . Initially, the wizard places all cards face down in a row, in unknown order.

You may repeatedly make moves of the following form: you point to any k of the cards. The wizard then turns those cards face up. If any two of the cards match, the game is over and you win. Otherwise, you must look away, while the wizard arbitrarily permutes the k chosen cards and turns them back face-down. Then, it is your turn again.

We say this game is *winnable* if there exist some positive integer m and some strategy that is guaranteed to win in at most m moves, no matter how the wizard responds.

For which values of n and k is the game winnable?

§46 USAMO 2017

1. (USAMO 2017) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Prove that there are infinitely many distinct pairs (a, b) of relatively prime positive integers $a > 1$ and $b > 1$ such that $a^b + b^a$ is divisible by $a + b$.

2. (USAMO 2017) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $m_1, m_2, \dots, m_n$ be a collection of n positive integers, not necessarily distinct. For any sequence of integers $A = (a_1, \dots, a_n)$ and any permutation $w = w_1, \dots, w_n$ of $m_1, \dots, m_n$, define an A -inversion of w to be a pair of entries w_i, w_j with $i < j$ for which one of the following conditions holds:

$$a_i \geq w_i > w_j,$$

$$w_j > a_i \geq w_i,$$

or

$$w_i > w_j > a_i.$$

Show that, for any two sequences of integers $A = (a_1, \dots, a_n)$ and $B = (b_1, \dots, b_n)$, and for any positive integer k , the number of permutations of $m_1, \dots, m_n$ having exactly k A -inversions is equal to the number of permutations of $m_1, \dots, m_n$ having exactly k B -inversions.

3. (*USAMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let ABC be a scalene triangle with circumcircle Ω and incenter I . Ray AI meets $\overline{BC}$ at D and meets Ω again at M ; the circle with diameter $\overline{DM}$ cuts Ω again at K . Lines MK and BC meet at S , and N is the midpoint of $\overline{IS}$. The circumcircles of $\triangle KID$ and $\triangle MAN$ intersect at points L_1 and L_2 . Prove that Ω passes through the midpoint of either $\overline{IL_1}$ or $\overline{IL_2}$.

4. (*USAMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $P_1, P_2, \dots, P_{2n}$ be $2n$ distinct points on the unit circle $x^2 + y^2 = 1$, other than $(1, 0)$. Each point is colored either red or blue, with exactly n red points and n blue points. Let $R_1, R_2, \dots, R_n$ be any ordering of the red points. Let B_1 be the nearest blue point to R_1 traveling counterclockwise around the circle starting from R_1 . Then let B_2 be the nearest of the remaining blue points to R_2 travelling counterclockwise around the circle from R_2 , and so on, until we have labeled all of the blue points $B_1, \dots, B_n$. Show that the number of counterclockwise arcs of the form $R_i \rightarrow B_i$ that contain the point $(1, 0)$ is independent of the way we chose the ordering $R_1, \dots, R_n$ of the red points.

5. (*USAMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $\mathbf{Z}$ denote the set of all integers. Find all real numbers $c > 0$ such that there exists a labeling of the lattice points $(x, y) \in \mathbf{Z}^2$ with positive integers for which: only finitely many distinct labels occur, and for each label i , the distance between any two points labeled i is at least c^i .

6. (*USAMO 2017*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Find the minimum possible value of

$$\frac{a}{b^3 + 4} + \frac{b}{c^3 + 4} + \frac{c}{d^3 + 4} + \frac{d}{a^3 + 4}$$

given that a, b, c, d are nonnegative real numbers such that $a + b + c + d = 4$.

§47 USAMO 2018

1. (*USAMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let a, b, c be positive real numbers such that $a + b + c = 4\sqrt[3]{abc}$. Prove that

$$2(ab + bc + ca) + 4\min(a^2, b^2, c^2) \geq a^2 + b^2 + c^2.$$

2. (*USAMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f\left(x + \frac{1}{y}\right) + f\left(y + \frac{1}{z}\right) + f\left(z + \frac{1}{x}\right) = 1$$

for all $x, y, z > 0$ with $xyz = 1$.

3. (*USAMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

For a given integer $n \geq 2$, let $\{a_1, a_2, \dots, a_m\}$ be the set of positive integers less than n that are relatively prime to n . Prove that if every prime that divides m also divides n , then $a_1^k + a_2^k + \dots + a_m^k$ is divisible by m for every positive integer k .

4. (*USAMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let p be a prime, and let $a_1, \dots, a_p$ be integers. Show that there exists an integer k such that the numbers

$$a_1 + k, a_2 + 2k, \dots, a_p + pk$$

produce at least $\frac{1}{2}p$ distinct remainders upon division by p .

5. (*USAMO 2018*) Note: For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

In convex cyclic quadrilateral $ABCD$, we know that lines AC and BD intersect at E , lines AB and CD intersect at F , and lines BC and DA intersect at G . Suppose that the circumcircle of $\triangle ABE$ intersects line CB at B and P , and the circumcircle of $\triangle ADE$ intersects line CD at D and Q , where C, B, P, G and C, Q, D, F are collinear in that order. Prove that if lines FP and GQ intersect at M , then $\angle MAC = 90^\circ$.

6. (*USAMO 2018*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let a_n be the number of permutations $(x_1, x_2, \dots, x_n)$ of the numbers $(1, 2, \dots, n)$ such that the n ratios $\frac{x_k}{k}$ for $1 \leq k \leq n$ are all distinct. Prove that a_n is odd for all $n \geq 1$.

§48 USAMO 2019

1. (*USAMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $\mathbb{N}$ be the set of positive integers. A function $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfies the equation

$$\underbrace{f(f(\dots f(n)\dots))}_{f(n) \text{ times}} = \frac{n^2}{f(f(n))}$$

for all positive integers n . Given this information, determine all possible values of $f(1000)$.

2. (*USAMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $ABCD$ be a cyclic quadrilateral satisfying $AD^2 + BC^2 = AB^2$. The diagonals of $ABCD$ intersect at E . Let P be a point on side $\overline{AB}$ satisfying $\angle APD = \angle BPC$. Show that line PE bisects $\overline{CD}$.

3. (*USAMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let K be the set of all positive integers that do not contain the digit 7 in their base-10 representation. Find all polynomials f with nonnegative integer coefficients such that $f(n) \in K$ whenever $n \in K$.

4. (*USAMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let n be a nonnegative integer. Determine the number of ways that one can choose $(n+1)^2$ sets $S_{i,j} \subseteq \{1, 2, \dots, 2n\}$, for integers i, j with $0 \leq i, j \leq n$, such that: for all $0 \leq i, j \leq n$, the set $S_{i,j}$ has $i+j$ elements; and $S_{i,j} \subseteq S_{k,l}$ whenever $0 \leq i \leq k \leq n$ and $0 \leq j \leq l \leq n$.

5. (*USAMO 2019*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled

diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Two rational numbers $\frac{m}{n}$ and $\frac{n}{m}$ are written on a blackboard, where m and n are relatively prime positive integers. At any point, Evan may pick two of the numbers x and y written on the board and write either their arithmetic mean $\frac{x+y}{2}$ or their harmonic mean $\frac{2xy}{x+y}$ on the board as well. Find all pairs (m, n) such that Evan can write 1 on the board in finitely many steps.

6. (USAMO 2019) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Find all polynomials P with real coefficients such that

$$\frac{P(x)}{yz} + \frac{P(y)}{zx} + \frac{P(z)}{xy} = P(x-y) + P(y-z) + P(z-x)$$

holds for all nonzero real numbers x, y, z satisfying $2xyz = x + y + z$.

§49 USAMO 2020

- (USAMO 2020) Let ABC be a fixed acute triangle inscribed in a circle ω with center O . A variable point X is chosen on minor arc AB of ω , and segments CX and AB meet at D . Denote by O_1 and O_2 the circumcenters of triangles ADX and BDX , respectively. Determine all points X for which the area of triangle OO_1O_2 is minimized.
- (USAMO 2020) An empty $2020 \times 2020 \times 2020$ cube is given, and a 2020×2020 grid of square unit cells is drawn on each of its six faces. A beam is a $1 \times 1 \times 2020$ rectangular prism. Several beams are placed inside the cube subject to the following conditions:
 - The two 1×1 faces of each beam coincide with unit cells lying on opposite faces of the cube. (Hence, there are $3 \cdot 2020^2$ possible positions for a beam.)
 - No two beams have intersecting interiors.
 - The interiors of each of the four 1×2020 faces of each beam touch either a face of the cube or the interior of the face of another beam.

What is the smallest positive number of beams that can be placed to satisfy these conditions?

- (USAMO 2020) Let p be an odd prime. An integer x is called a *quadratic non-residue* if p does not divide $x - t^2$ for any integer t . Denote by A the set of all integers a such that $1 \leq a < p$, and both a and $4 - a$ are quadratic non-residues. Calculate the remainder when the product of the elements of A is divided by p .
- (USAMO 2020) Suppose that $(a_1, b_1), (a_2, b_2), \dots, (a_{100}, b_{100})$ are distinct ordered pairs of nonnegative integers. Let N denote the number of pairs of integers (i, j) satisfying $1 \leq i < j \leq 100$ and $|a_i b_j - a_j b_i| = 1$. Determine the largest possible value of N over all possible choices of the 100 ordered pairs.

5. (*USAMO 2020*) A finite set S of points in the coordinate plane is called *overdetermined* if $|S| \geq 2$ and there exists a nonzero polynomial $P(t)$, with real coefficients and of degree at most $|S| - 2$, satisfying $P(x) = y$ for every point $(x, y) \in S$.

For each integer $n \geq 2$, find the largest integer k (in terms of n) such that there exists a set of n distinct points that is not overdetermined, but has k overdetermined subsets.

6. (*USAMO 2020*) Let $n \geq 2$ be an integer. Let $x_1 \geq x_2 \geq \cdots \geq x_n$ and $y_1 \geq y_2 \geq \cdots \geq y_n$ be $2n$ real numbers such that

$$0 = x_1 + x_2 + \cdots + x_n = y_1 + y_2 + \cdots + y_n$$

$$\text{and } 1 = x_1^2 + x_2^2 + \cdots + x_n^2 = y_1^2 + y_2^2 + \cdots + y_n^2.$$

Prove that

$$\sum_{i=1}^n (x_i y_i - x_i y_{n+1-i}) \geq \frac{2}{\sqrt{n-1}}.$$

§50 USAMO 2021

1. (*USAMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Rectangles BCC_1B_2 , CAA_1C_2 , and ABB_1A_2 are erected outside an acute triangle ABC . Suppose that

$$\angle BC_1C + \angle CA_1A + \angle AB_1B = 180^\circ.$$

Prove that lines B_1C_2 , C_1A_2 , and A_1B_2 are concurrent.

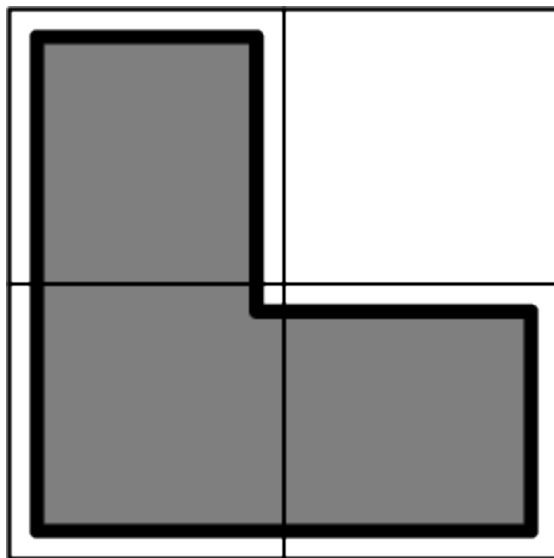
2. (*USAMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

The Planar National Park is a subset of the Euclidean plane consisting of several trails which meet at junctions. Every trail has its two endpoints at two different junctions whereas each junction is the endpoint of exactly three trails. Trails only intersect at junctions (in particular, trails only meet at endpoints). Finally, no trails begin and end at the same two junctions.

A visitor walks through the park as follows: she begins at a junction and starts walking along a trail. At the end of that first trail, she enters a junction and turns left. On the next junction she turns right, and so on, alternating left and right turns at each junction. She does this until she gets back to the junction where she started. What is the largest possible number of times she could have entered any junction during her walk, over all possible layouts of the park?

3. (*USAMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $n \geq 2$ be an integer. An $n \times n$ board is initially empty. Each minute, you may perform one of three moves: If there is an L-shaped tromino region of three cells without stones on the board (see figure; rotations not allowed), you may place a stone in each of those cells. If all cells in a column have a stone, you may remove all stones from that column. If all cells in a row have a stone, you may remove all stones from that row.



For which n is it possible that, after some non-zero number of moves, the board has no stones?

4. (*USAMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

A finite set S of positive integers has the property that, for each $s \in S$, and each positive integer divisor d of s , there exists a unique element $t \in S$ satisfying $\gcd(s, t) = d$. (The elements s and t could be equal.)

Given this information, find all possible values for the number of elements of S .

5. (*USAMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

Let $n \geq 4$ be an integer. Find all positive real solutions to the following system of $2n$ equations:

$$a_1 = \frac{1}{a_{2n}} + \frac{1}{a_2}, \quad a_2 = a_1 + a_3,$$

$$\begin{aligned}
 a_3 &= \frac{1}{a_2} + \frac{1}{a_4}, & a_4 &= a_3 + a_5, \\
 a_5 &= \frac{1}{a_4} + \frac{1}{a_6}, & a_6 &= a_5 + a_7, \\
 &\vdots & &\vdots \\
 a_{2n-1} &= \frac{1}{a_{2n-2}} + \frac{1}{a_{2n}}, & a_{2n} &= a_{2n-1} + a_1
 \end{aligned}$$

6. (*USAMO 2021*) **Note:** For any geometry problem whose statement begins with an asterisk (*), the first page of the solution must be a large, in-scale, clearly labeled diagram. Failure to meet this requirement will result in an automatic 1-point deduction.

(*) Let $ABCDEF$ be a convex hexagon satisfying $\overline{AB} \parallel \overline{DE}$, $\overline{BC} \parallel \overline{EF}$, $\overline{CD} \parallel \overline{FA}$, and

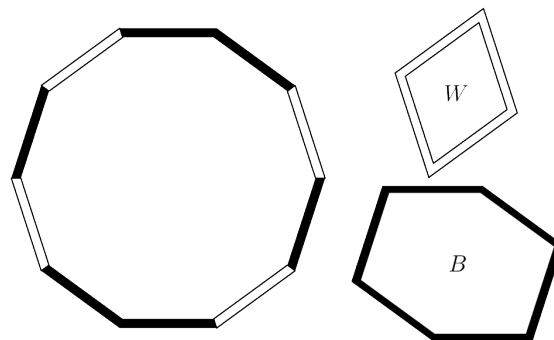
$$AB \cdot DE = BC \cdot EF = CD \cdot FA.$$

Let X , Y , and Z be the midpoints of $\overline{AD}$, $\overline{BE}$, and $\overline{CF}$. Prove that the circumcenter of $\triangle ACE$, the circumcenter of $\triangle BDF$, and the orthocenter of $\triangle XYZ$ are collinear.

§51 USAMO 2022

1. (*USAMO 2022*) Let a and b be positive integers. The cells of an $(a+b+1) \times (a+b+1)$ grid are colored amber and bronze such that there are at least $a^2 + ab - b$ amber cells and at least $b^2 + ab - a$ bronze cells. Prove that it is possible to choose a amber cells and b bronze cells such that no two of the $a+b$ chosen cells lie in the same row or column.
2. (*USAMO 2022*) Let $b \geq 2$ and $w \geq 2$ be fixed integers, and $n = b + w$. Given are $2b$ identical black rods and $2w$ identical white rods, each of side length 1.

We assemble a regular $2n$ -gon using these rods so that parallel sides are the same color. Then, a convex $2b$ -gon B is formed by translating the black rods, and a convex $2w$ -gon W is formed by translating the white rods. An example of one way of doing the assembly when $b = 3$ and $w = 2$ is shown below, as well as the resulting polygons B and W .



Prove that the difference of the areas of B and W depends only on the numbers b and w , and not on how the $2n$ -gon was assembled.

3. (USAMO 2022) Let $\mathbb{R}_{>0}$ be the set of all positive real numbers. Find all functions $f : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ such that for all $x, y \in \mathbb{R}_{>0}$ we have

$$f(x) = f(f(f(x)) + y) + f(xf(y))f(x + y).$$

4. (USAMO 2022) Find all pairs of primes (p, q) for which $p - q$ and $pq - q$ are both perfect squares.
5. (USAMO 2022) A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *essentially increasing* if $f(s) \leq f(t)$ holds whenever $s \leq t$ are real numbers such that $f(s) \neq 0$ and $f(t) \neq 0$.

Find the smallest integer k such that for any 2022 real numbers $x_1, x_2, \dots, x_{2022}$, there exist k essentially increasing functions $f_1, \dots, f_k$ such that

$$f_1(n) + f_2(n) + \dots + f_k(n) = x_n \quad \text{for every } n = 1, 2, \dots, 2022.$$

6. (USAMO 2022) There are 2022 users on a social network called Mathbook, and some of them are Mathbook-friends. (On Mathbook, friendship is always mutual and permanent.)

Starting now, Mathbook will only allow a new friendship to be formed between two users if they have *at least two* friends in common. What is the minimum number of friendships that must already exist so that every user could eventually become friends with every other user?

§52 USAMO 2023

1. (USAMO 2023) In an acute triangle ABC , let M be the midpoint of $\overline{BC}$. Let P be the foot of the perpendicular from C to AM . Suppose the circumcircle of triangle ABP intersects line BC at two distinct points B and Q . Let N be the midpoint of $\overline{AQ}$. Prove that $NB = NC$.
2. (USAMO 2023) Let $\mathbb{R}^+$ be the set of positive real numbers. Find all functions $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that, for all $x, y \in \mathbb{R}^+$,

$$f(xy + f(x)) = xf(y) + 2$$

3. (USAMO 2023) Consider an n -by- n board of unit squares for some odd positive integer n . We say that a collection C of identical dominoes is a maximal grid-aligned configuration on the board if C consists of $(n^2 - 1)/2$ dominoes where each domino covers exactly two neighboring squares and the dominoes don't overlap: C then covers all but one square on the board. We are allowed to slide (but not rotate) a domino on the board to cover the uncovered square, resulting in a new maximal grid-aligned configuration with another square uncovered. Let $k(C)$ be the number of distinct maximal grid-aligned configurations obtainable from C by repeatedly sliding dominoes. Find all possible values of $k(C)$ as a function of n .
4. (USAMO 2023) A positive integer a is selected, and some positive integers are written on a board. Alice and Bob play the following game. On Alice's turn, she must replace some integer n on the board with $n + a$, and on Bob's turn he must replace some even integer n on the board with $n/2$. Alice goes first and they alternate turns. If on his turn Bob has no valid moves, the game ends.

After analyzing the integers on the board, Bob realizes that, regardless of what moves Alice makes, he will be able to force the game to end eventually. Show that, in fact, for this value of a and these integers on the board, the game is guaranteed to end regardless of Alice's or Bob's moves.

5. (USAMO 2023) Let $n \geq 3$ be an integer. We say that an arrangement of the numbers $1, 2, \dots, n^2$ in a $n \times n$ table is row-valid if the numbers in each row can be permuted to form an arithmetic progression, and column-valid if the numbers in each column can be permuted to form an arithmetic progression. For what values of n is it possible to transform any row-valid arrangement into a column-valid arrangement by permuting the numbers in each row?
6. (USAMO 2023) Let ABC be a triangle with incenter I and excenters I_a, I_b, I_c opposite A, B , and C , respectively. Given an arbitrary point D on the circumcircle of $\triangle ABC$ that does not lie on any of the lines II_a, I_bI_c , or BC , suppose the circumcircles of $\triangle DII_a$ and $\triangle DI_bI_c$ intersect at two distinct points D and F . If E is the intersection of lines DF and BC , prove that $\angle BAD = \angle EAC$.

§53 USAMO 2024

1. (USAMO 2024) Find all integers $n \geq 3$ such that the following property holds: if we list the divisors of $n!$ in increasing order as $1 = d_1 < d_2 < \dots < d_k = n!$, then we have

$$d_2 - d_1 \leq d_3 - d_2 \leq \dots \leq d_k - d_{k-1}.$$

2. (USAMO 2024) Let $S_1, S_2, \dots, S_{100}$ be finite sets of integers whose intersection is not empty. For each non-empty $T \subseteq \{S_1, S_2, \dots, S_{100}\}$, the size of the intersection of the sets in T is a multiple of the number of sets in T . What is the least possible number of elements that are in at least 50 sets?
3. (USAMO 2024) Let m be a positive integer. A triangulation of a polygon is m -balanced if its triangles can be colored with m colors in such a way that the sum of the areas of all triangles of the same color is the same for each of the m colors. Find all positive integers n for which there exists an m -balanced triangulation of a regular n -gon.

Note: A triangulation of a convex polygon $\mathcal{P}$ with $n \geq 3$ sides is any partitioning of $\mathcal{P}$ into $n - 2$ triangles by $n - 3$ diagonals of $\mathcal{P}$ that do not intersect in the polygon's interior.

4. (USAMO 2024) Let m and n be positive integers. A circular necklace contains mn beads, each either red or blue. It turned out that no matter how the necklace was cut into m blocks of n consecutive beads, each block had a distinct number of red beads. Determine, with proof, all possible values of the ordered pair (m, n) .
5. (USAMO 2024) Point D is selected inside acute triangle ABC so that $\angle DAC = \angle ACB$ and $\angle BDC = 90^\circ + \angle BAC$. Point E is chosen on ray BD so that $AE = EC$. Let M be the midpoint of BC . Show that line AB is tangent to the circumcircle of triangle BEM .
6. (USAMO 2024) Let $n > 2$ be an integer and let $\ell \in \{1, 2, \dots, n\}$. A collection $A_1, \dots, A_k$ of (not necessarily distinct) subsets of $\{1, 2, \dots, n\}$ is called ℓ -large if

$|A_i| \geq \ell$ for all $1 \leq i \leq k$. Find, in terms of n and ℓ , the largest real number c such that the inequality

$$\sum_{i=1}^k \sum_{j=1}^k x_i x_j \frac{|A_i \cap A_j|^2}{|A_i| \cdot |A_j|} \geq c \left(\sum_{i=1}^k x_i \right)^2$$

holds for all positive integers k , all nonnegative real numbers $x_1, \dots, x_k$, and all ℓ -large collections $A_1, \dots, A_k$ of subsets of $\{1, 2, \dots, n\}$.

Note: For a finite set S , $|S|$ denotes the number of elements in S .

§54 USAMO 2025

- (USAMO 2025) Let k and d be positive integers. Prove that there exists a positive integer N such that for every odd integer $n > N$, the digits in the base-2 representation of n^k are all greater than d .
- (USAMO 2025) Let n and k be positive integers with $k < n$. Let $P(x)$ be a polynomial of degree n with real coefficients, nonzero constant term, and no repeated roots. Suppose that for any real numbers $a_0, a_1, \dots, a_k$ such that the polynomial $a_k x^k + \dots + a_1 x + a_0$ divides $P(x)$, the product $a_0 a_1 \dots a_k$ is zero. Prove that $P(x)$ has a nonreal root.
- (USAMO 2025) Alice the architect and Bob the builder play a game. First, Alice chooses two points P and Q in the plane and a subset $\mathcal{S}$ of the plane, which are announced to Bob. Next, Bob marks infinitely many points in the plane, designating each a city. He may not place two cities within distance at most one unit of each other, and no three cities he places may be collinear. Finally, roads are constructed between the cities as follows: for each pair A, B of cities, they are connected with a road along the line segment AB if and only if the following condition holds: For every city C distinct from A and B , there exists $R \in \mathcal{S}$ such that $\triangle PQR$ is directly similar to either $\triangle ABC$ or $\triangle BAC$. Alice wins the game if (i) the resulting roads allow for travel between any pair of cities via a finite sequence of roads and (ii) no two roads cross. Otherwise, Bob wins. Determine, with proof, which player has a winning strategy.

Note: $\triangle UVW$ is directly similar to $\triangle XYZ$ if there exists a sequence of rotations, translations, and dilations sending U to X , V to Y , and W to Z .

- (USAMO 2025) Let H be the orthocenter of acute triangle ABC , let F be the foot of the altitude from C to AB , and let P be the reflection of H across BC . Suppose that the circumcircle of triangle AFP intersects line BC at two distinct points X and Y . Prove that C is the midpoint of XY .
- (USAMO 2025) Determine, with proof, all positive integers k such that

$$\frac{1}{n+1} \sum_{i=0}^n \binom{n}{i}^k$$

is an integer for every positive integer n .

- (USAMO 2025) Let m and n be positive integers with $m \geq n$. There are m cupcakes of different flavors arranged around a circle and n people who like cupcakes. Each

person assigns a nonnegative real number score to each cupcake, depending on how much they like the cupcake. Suppose that for each person P , it is possible to partition the circle of m cupcakes into n groups of consecutive cupcakes so that the sum of P 's scores of the cupcakes in each group is at least 1. Prove that it is possible to distribute the m cupcakes to the n people so that each person P receives cupcakes of total score at least 1 with respect to P .

§55 USAMO 2026

1. (*USAMO 2026*) Let n be an integer greater than 1. For which real numbers x is

$$\lfloor nx \rfloor - \sum_{k=1}^n \frac{\lfloor kx \rfloor}{k}$$

maximal, and what is the maximal value that this expression can take? Note: $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z .

2. (*USAMO 2026*) Annie is playing a game where she starts with a row of positive integers, written on a blackboard, each of which is a power of 2. On each turn, she can erase two adjacent numbers and replace them with a power of 2 that is greater than either of the erased numbers. This shortens the row of numbers, and she continues to take turns until only one number remains. Annie wins the game if the final remaining number is less than 4 times the sum of the original numbers. Is it always possible for Annie to win, regardless of the starting row of numbers?

3. (*USAMO 2026*) Let ABC be an acute scalene triangle with no angle equal to 60° . Let ω be the circumcircle of ABC . Let $\triangle_B$ be the equilateral triangle with three vertices on ω , one of which is B . Let ℓ_B be the line through the two vertices of $\triangle_B$ other than B . Let $\triangle_C$ and ℓ_C be defined analogously. Let Y be the intersection of AC and ℓ_B , and let Z be the intersection of AB and ℓ_C .

Let N be the midpoint of minor arc BC on ω . Let $\mathcal{R}$ be the triangle formed by ℓ_B , ℓ_C , and the tangent to ω through N . Prove that the circumcircle of AYZ and the incircle of $\mathcal{R}$ are tangent.

4. (*USAMO 2026*) A positive integer n is called solitary if, for any nonnegative integers a and b such that $a + b = n$, either a or b contains the digit 1 when written in base 10. Determine, with proof, the number of solitary integers less than 10^{2026} .
5. (*USAMO 2026*) Let ABC be a triangle. Points D , E , and F lie on sides BC , CA , and AB , respectively, such that

$$\angle AFE = \angle BDF = \angle CED.$$

Let O_A , O_B , and O_C be the circumcenters of triangles AFE , BDF , and CED , respectively. Let M , N , and O be the circumcenters of triangles ABC , DEF , and $O_AO_BO_C$, respectively. Prove that $OM = ON$.

6. (*USAMO 2026*) Let a and b be positive integers such that $\varphi(ab+1)$ divides a^2+b^2+1 . Prove that a and b are Fibonacci numbers.

Note: Euler's totient function $\varphi(n)$ is the number of positive integers less than or equal to n that are relatively prime to n .